

Conversational Cache Retrieval-Augmented Fine-Tuning System for Diagnosis of Learning Disabilities in Children

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ABSTRACT The early detection of learning disabilities (LDs) is critical for providing timely interventions and support to children facing academic challenges. However, a shortage of psychologists and the complexity of learning disability diagnostics often lead to delays. This paper proposes a Conversational Cache Retrieval-Augmented Fine-Tuning system designed to assist psychologists in diagnosing LDs, specifically targeting Auditory Processing Disorder (APD), Autism Spectrum Disorder (ASD), Attention Deficit Hyperactivity Disorder (ADHD), Dyslexia, and Dyscalculia. The system leverages artificial intelligence (AI), retrieval-augmented generation (RAG) and fine-tuning pre-trained models on datasets relevant to LDs to provide personalized diagnostic conversations with children, offering tailored questions based on real-time context retrieval from a large corpus of clinical data. The fine-tuning process allows the model to adapt to the nuances of learning disability diagnosis, enhancing its ability to generate accurate insights and reports. User input is collected through structured conversations inspired by established screening tests such as the Denver Developmental Screening Test and the Ages and Stages Questionnaire. The system then analyzes the input and generates a report, identifying possible red flags for psychologists to review. Through rigorous testing, the system demonstrates high precision in identifying LDs, providing a scalable and efficient tool to accelerate the diagnostic process and improve access to care in underserved areas.

INDEX TERMS Chat-bot, Conversational Artificial Intelligence, Fine-tuning, Learning Disabilities, Machine Learning, Retrieval-Augmented Generation

I. INTRODUCTION

LEARNING disabilities (LD) are neuro-developmental disorders that affect the ability to learn and perform simple tasks. A child with a learning disability may have difficulty reading, writing, speaking, listening, understanding mathematics and general comprehension [1]. They also struggle with a complex range of internal and external challenges. Internally, they may grapple with intense feelings of anxiety and depression [2]. Externally, they may confront difficulties associated with socialization and misconduct and poor educational performance [1] [2]. Learning disabilities can co-occur with other neuro-developmental disorders, this is defined as co-morbidity. For instance, the co-occurrence of conditions such as attention deficit disorder (ADD) alongside autism spectrum disorder (ASD) and dyslexia.

Traditional means of diagnosing learning disabilities rely on mental assessments done by psychologists. However,

these methods are time consuming and require a large number of psychologists who are not always available [3] [4]. A study with similar focus has shown that advancements in machine learning, particularly conversational artificial intelligence (AI), finetuning, and retrieval-augmented generation (RAG) have opened avenues for addressing these challenges [5]. Machine learning is the development of systems that can learn without being explicitly coded. By using these advancements, AI models can serve as support tools to psychologists in diagnosing learning disabilities.

Conversational AI is a field in machine learning that leverages natural language processing to simulate human-like conversations. It can be applied to detect signs of learning disabilities by asking questions, providing responses [6] and performing reasoning on learning disability related queries [7] [4] [8]. Additionally, it can generate a report that can assist psychologists in making a diagnostic summary

[9]. However, for a conversational AI model to be effective in learning disability detection, it needs to be adapted to the task. This can be done through finetuning and retrieval augmented generation.

Finetuning leverages the pretrained model's existing knowledge to adapt it to learning disability detection. The process involves training the model on smaller, more specific datasets such as clinical trial records and research studies relevant to learning disabilities. Conversational AI models typically generate generalised responses. Fine-tuning refines them for specific diagnostic tasks, [6] [10]. Compared to training a model from scratch, fine-tuning is more cost efficient, reduces training time and conserves computational resources by utilizing prior knowledge from the base model [11]. Additionally, conversational AI suffers from hallucinations-where it gives wrong answers because of its probabilistic nature and knowledge obsolescence-where the previously acquired knowledge becomes outdated. Retrieval-Augmented Generation mitigates this by integrating retrieval mechanisms [6] to retrieve up-to-date information relevant to learning disability diagnosis so that the model provides contextually accurate responses [12] and ask up to standard questions when querying users on the potential signs of learning disabilities in a child.

One implementation of conversational AI is chatbots. Chatbots are computer programs designed to simulate conversation with human users. With the integration of machine learning techniques like fine tuning using datasets relevant to learning disabilities and retrieval augmented generation, the chatbots can be fitted to ask questions relevant to learning disability detection. The user input collected is analysed and thereafter a report that psychologists can use to generate a synopsis is generated [6]. With this approach, the time taken to diagnose a child is shortened. Also, because these machine learning approaches are widely available on various digital platforms, the service is accessible by every individual that experiences a shortage of psychologists in their area.

This work aims to explore the integration of conversational AI, retrieval-augmented generation, and finetuning to develop a chatbot aimed at supporting psychologists in LD diagnosis. The conversational AI model is trained on datasets relevant to learning disabilities to provide users with intelligent, tailored responses such as follow up questions on the indication of learning disabilities. Additionally, the retrieval-augmented approach ensures that the generated responses are grounded in accurate, up-to-date information.

II. LITERATURE REVIEW

Machine learning has been increasingly applied in early detection and diagnosis of learning disabilities such as ADHD, Dyslexia and Dyscalculia. [13] Recent advancements in natural language processing (NLP), deep learning and retrieval-based models have enabled development of chatbots that can assess cognitive and linguistic patterns in children. [14] Supervised learning techniques have been extensively employed to classify and predict learning disabilities. Decision trees

and random forests have been utilized to predict learning disorders in pediatric populations, displaying the potential of these models in early detection efforts [13]. Similarly, a fusion of Naïve Bayes and neural network classifiers has been applied to predict learning disabilities in school children, achieving notable accuracy [15]. NLP techniques are pivotal in analyzing linguistic patterns associated with learning disabilities. Studies have employed machine learning models to detect autism spectrum disorders by analyzing speech transcripts, highlighting the efficacy of NLP in identifying atypical language use. NLP techniques are pivotal in analyzing linguistic patterns associated with learning disabilities. Studies have employed machine learning models to detect autism spectrum disorders by analyzing speech transcripts, highlighting the efficacy of NLP in identifying atypical language use. [16] Additionally, research has demonstrated the potential of classifying autism from semi-structured speech recordings using ML approaches, underscoring the role of speech analysis in early diagnosis. [17] The integration of ML algorithms into chatbots offers a dynamic tool for assessing learning disabilities. AI-driven adaptive learning platforms have been developed to enhance education for students with disabilities, providing personalized support and facilitating the detection of learning challenges. [17] Moreover, AI applications such as chat robots and intelligent tutors have been employed to support students with learning disabilities, demonstrating the versatility of chatbots in educational settings. [18] [19] A significant challenge in developing ML-based diagnostic tools is the scarcity of high-quality, labeled datasets. Many existing datasets are limited in size and diversity, which can impede the training of robust models. [16] [19] To mitigate this, researchers have explored data augmentation techniques and the use of synthetic data to enhance model performance. [15] Ensuring that ML models are interpretable and free from biases related to demographic factors is crucial. The deployment of AI in educational contexts necessitates transparency to build trust among users and stakeholders. [18] Additionally, ethical considerations, such as avoiding over-reliance on AI and protecting student privacy, are paramount. [20] Future research should focus on integrating multimodal data—including text, speech, and behavioral metrics—to enhance the accuracy of learning disability detection. Combining various data sources can provide a more comprehensive understanding of a child's learning profile. [21] Implementing ML-based chatbots in real-world educational settings requires continuous monitoring and updating of models to adapt to diverse student populations. Collaborations between AI researchers, educators, and clinicians are essential to develop tools that are both effective and ethically sound.

III. PROBLEM STATEMENT

Learning disabilities can lead to long term problems such as poor mental health, unemployment, and socialization and misconduct issues. Despite the growing recognition of learning disabilities as a critical aspect of overall well-being, many

individuals continue to face barriers in accessing timely and effective care. Particularly in regions with limited mental healthcare infrastructure, where the shortage of psychologists exacerbates the issue. As a result, these services are costly and therefore inaccessible to most children. Additionally, screening test such as the Denver Developmental Screening Test and Ages and Stage Questionnaire that are commonly used to identify learning disabilities require follow-up analysis by psychologists. This can cause delays in diagnosis especially in resource-limited regions.

However, some of the primary challenges in integrating machine learning into mental health services include ethical concerns surrounding data usage and patient confidentiality [4] [22] [23]. These systems rely on vast amounts of personal data to function such as sensitive mental health records which raises concerns about the privacy and potential misuse of such data [24] [25]. Inadvertently, models may reinforce biasness present in data used to train them [5]. Further more, there needs to be mechanisms in place to defend against malicious inputs (adversarial attacks) [26] aimed at confusing or exploiting the system which can lower its reliability and accuracy and later cause a misdiagnosis.

Additionally, retrieving relevant information from large knowledge bases related to learning disabilities can be difficult (high dimensionality). This can cause high computational overhead because of the need to balance performance with resource techniques. Accurately identifying and linking resources to identify signs of the disorders across different sources may present a problem, failure to do so accurately may create a bias in during the report generation of the system. Some systems present themselves as black boxes and provide little to no understanding of the reasoning behind their decisions. This can cause resistance by psychologists to adopt the approach as a support tool when diagnosing learning disabilities [5] [24]. Maintaining conversational flow that is both informative and supportive for learning disabilities can be difficult.

IV. PROPOSED SOLUTION

A conversational retrieval-augmented fine-tuned system for early detection of learning disabilities can be a scalable and cost-effective solution to the scarcity of mental health personnel [3] [24]. Using conversational artificial intelligence through a chatbot that uses natural language processing to engage in human-like conversations [27] [28], understand user queries, and provide relevant contextual responses to collect user data, and ask questions relevant learning disabilities [7] [4] [8]. However, conversational AI suffers from hallucinations (where it generates confident but incorrect statements) [6] and knowledge obsolescence [9]. Retrieval-Augmented Generation mitigates these issues by integrating retrieval mechanisms to allow the chatbot to reference up-to-date verifiable information from mental health resources [5]. This allows it to ask questions relevant to learning disorders.

Retrieval augmented generation alone is not enough to ensure the accuracy of the chatbot. A base model is finetuned

on datasets relevant to learning disabilities to specialise the chatbot to tasks specific to learning disability detection [10] [29]. The chatbot can have a human-like conversation with a user to collect user input. The knowledge it retrieves using RAG is used to make sure the chatbot asks up-to-date questions [6] relevant to learning disability detection. To avoid generalised responses, the base model is pre-trained on datasets relevant to learning disabilities such as clinical trials to adapt it to the detection of the disorders. The result would be an automated learning disability assessment support tool that can be used to hasten their diagnosis.

Conversational artificial intelligence designed for on-device processing minimizes data exposure [30] and RAG can retrieve anonymized documents together, promoting data privacy and security. Fine-tuning uses adversarial training to recognize unusual input patterns [26], with real-time monitoring by conversational AI to detect and flag malicious activity and dissolve adversarial attacks. RAG can be used to fetch only relevant documents to reduce storage needs and high dimensionality [31]. RAG can retrieve real-time domain specific documents on learning disabilities alongside fine-tuning on learning disability datasets to ensure that complex medical terms are understood improving accuracy and decision support. RAG also increases transparency by providing citations for the retrieved knowledge. Conversational AI can continuously learn from underrepresented groups to mitigate bias.

V. DATASET DESCRIPTION

The dataset used in this study consists of documents from trusted medical resources like the World Health Organisation (WHO), Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5), and American Psychological Institution (APA) with over 10000 records related to learning disabilities. The dataset focuses on Attention-Deficit/Hyperactivity Disorder (ADHD), Autism Spectrum Disorder (ASD), dyslexia, dyscalculia, and Auditory Processing Disorder (APD). The data derived from clinical trials and research studies on learning disabilities is based on real-world data that can be refined to accurately train the model and create an external corpus used by retrieval augmented generation to provide context to the large language model. The dataset consists of over 100 records. Each record contains a question-answer pair where the user query based on symptoms acts as the user input and the corresponding model's output describes the learning disability defined in the query.

The dataset contains features such as causes (the factors contributing to each neuro-developmental disorder) and symptoms which are specific signs related to each learning disability. The data is structured to include a label field for each record that corresponds to the learning disability described for example ADHD, ASD. This label field serves as the ground truth model for evaluation of the performance metrics of the model. Before the data retrieved from the documents retrieved by web scraping techniques like Beuti-

TABLE 1. The number of records per disorder used to train the model

Disorder	Number of records
Attention-Deficit/Hyperactivity Disorder (ADHD)	30
Autism Spectrum Disorder (ASD)	48
Dyslexia	30
Dyscalculia	30
Auditory Processing Disorder (APD)	30

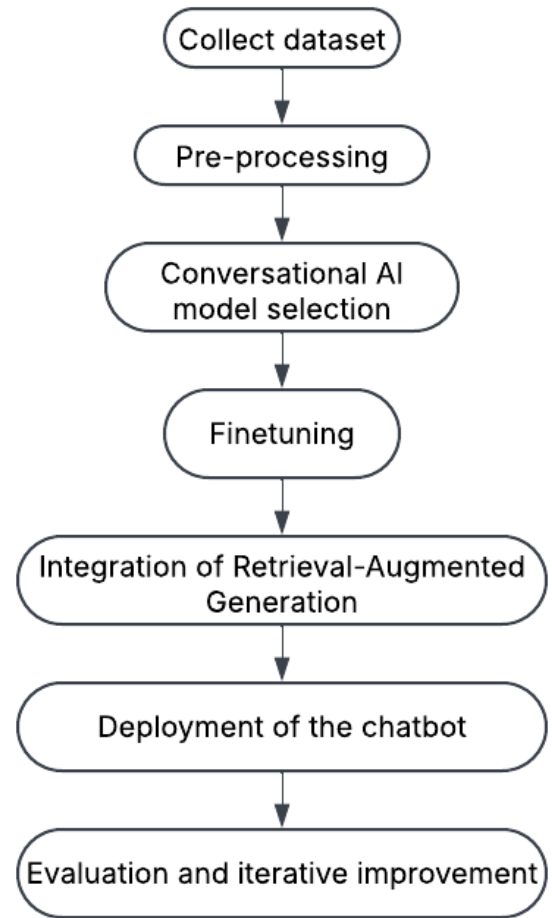
fulSoup are used, it is preprocessed. First, the documents are proofread, and cleaned to ensure accuracy, and relevancy by removing irrelevant or outdated information. Key information related to symptoms and causes of the learning disabilities is extracted from the documents. The extracted information is converted into question-answer format for ease of use in training the model. A structured dataset where user symptom-based queries are paired with the corresponding responses describing the learning disability is created. The dataset is carefully built to ensure balanced representation of the five learning disabilities. Equal attention was given to ensure that all learning disabilities had an adequate number of records to be used during training of the model. The dataset is distributed across the five learning disabilities as seen in table 1

All documents used in the dataset were publicly available from open-source, trusted medical resources. No personal, or identifiable patient data was included in the dataset, ensuring compliance with data standards. Data from clinical trials and research findings were used with the understanding that the information was freely available for research purposes. No proprietary or restricted data was accessed without proper permission.

VI. METHODOLOGY

To ensure accuracy, data is collected from a variety of trusted open-source mental health resources by scrapping documents on learning disabilities using web scraping tools like BeautifulSoup, Scrapy. Datasets compiled from related research studies are also used. However, before the data can be used, it needs to be cleaned. This is done by removing noise and duplicates that may occur due to overlapping sources. Missing data are handled by filling in missing values with standardized placeholders and text cleaning is done by converting text to a consistent case and removing punctuation. Next, the data is normalized and structured, numerical data is standardized to avoid inconsistencies and schema mapping is done to link key pieces of information to standardized fields.

After structuring the data, the extracted information is mapped to domain-specific vocabulary using vocabularies like UMLS to ensure consistency. The data is then validated by cross-checking it against trusted clinical trial registries and finally the cleaned data is exported in a format i.e. CSV that can be used in the fine-tuning stage to train the model to detect learning disabilities [29] [32]. There after retrieval augmented generation is integrated which allows for real-time retrieval of information relevant to learning disorders

**FIGURE 1.** The stages of the methodology including data collection, preprocessing, model development, and evaluation

so as to provide contextually accurate responses to learning disability identification. The chatbot system is then deployed and psychologists are allowed to interact with it to test its accuracy [33] [34]. The process is iterated to achieve the best possible outcome.

VII. EXPLORATORY DATA ANALYSIS (EDA)

Exploratory data analysis (EDA) is an important first step to analyse data. It involves looking at and visualizing data to understand its main features, find patterns, and discover how different parts of the data are connected. Before starting EDA, there is need of a clear understanding on the problem that needs to be solved. In this case, we're tackling the problem of delayed diagnosis of learning disabilities in children due to the shortage of psychologists. Exploratory data analysis was performed on the FLAMENCO dataset for learning disabilities to understand how autism and intellectual disability are related.

First, the data was loaded and examined to understand

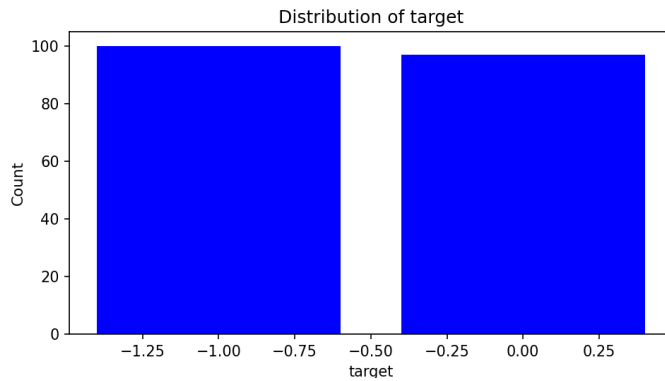


FIGURE 2. A bar chart to visualize the distribution of number of occurrences of the target

the number of features and observations present. There were 197 observations (rows) and fifteen features (columns). The columns present were caseId, ClientID, Verbal and Intellectual Ability, Verbalization, Syntax, Pragmatic Perception, Pre-writing, visual-motor integration, Sequencing, Memory, Perception, Recognition, Cognitive Flexibility, Attention deficit (Intellectual-ASD-ADHD), target. Then, the dataset was checked for missing values in every column. This dataset was complete and had no missing data. Next, the different datatypes for each variable were determined. The dataset was found to have one categorical column, target. The rest were numerical.

After inspecting the data and understanding its characteristics, the data was visualised using three strategies. The first strategy was univariate analysis to understand one variable at a time and its characteristics. For the categorical columns, bar charts were used during the visualisation. For numerical data, histograms were used to visualize their distribution. The mean, mode and median were calculated to understand the central spread of the data.

In figure 2 the bar chart illustrates the distribution of target labels in the FLAMENCO dataset. The x-axis represents the target categories and the y-axis the number of occurrences. There is a balanced distribution between the occurrences at around 100 instances.

In figure 3 the histogram that shows the distribution of verbalization across the dataset. It is right skewed because the mean is around 30 and is greater than the median and the mode. This shows that most individuals with learning disabilities perform poorly in understanding and using language in social context.

In figure 4 the Kernel Density Estimates show how data is distributed. For pragmatic perception, the distribution is left skewed suggesting that the density is concentrated towards higher values. This implies that most individuals perform well in using language for social context and the few that struggle may be learning disability victims.

Figure 5 shows the distribution of how the different categories within the target variables are spread across the

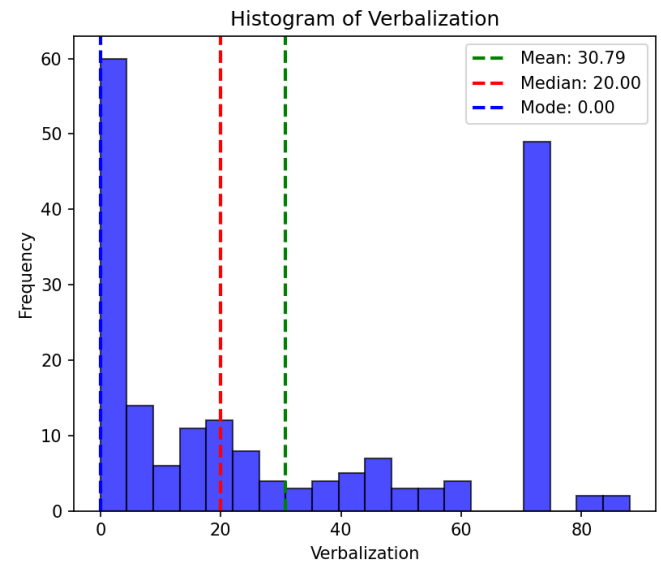


FIGURE 3. A histogram showing the visualization of verbalization and pre-writing

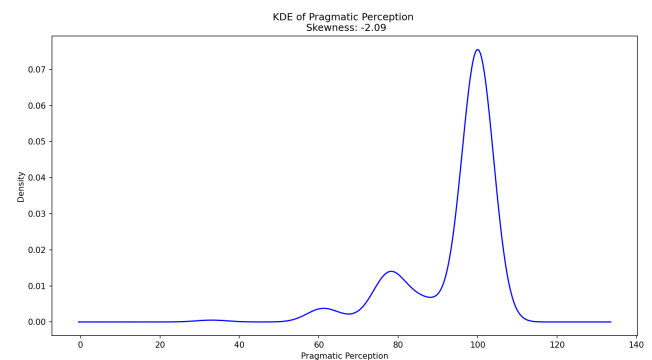


FIGURE 4. Kernel Density Estimate of different features

dataset. -1 represents individuals with learning disabilities while 0 represents those without. From the pie chart, it is observed that a greater percentage of individuals suffer with learning disabilities.

In figure 6 the X-axis in the violin plot shows scores attained by individuals when tested on a certain feature while the y-axis shows normalized values where 1.0 indicates the norm. The wide sections show the regions where most participants' scores lie while the narrow regions suggest fewer participants at the extreme. For verbal and intellectual ability, the violin plot is symmetrical indicating a normal distribution.

Figure 7 shows a density plot for pre-writing to illustrate where most data points are concentrated and the spread of data. The distribution is right skewed. With a peak at around 26. The individuals that scored more most probably do not

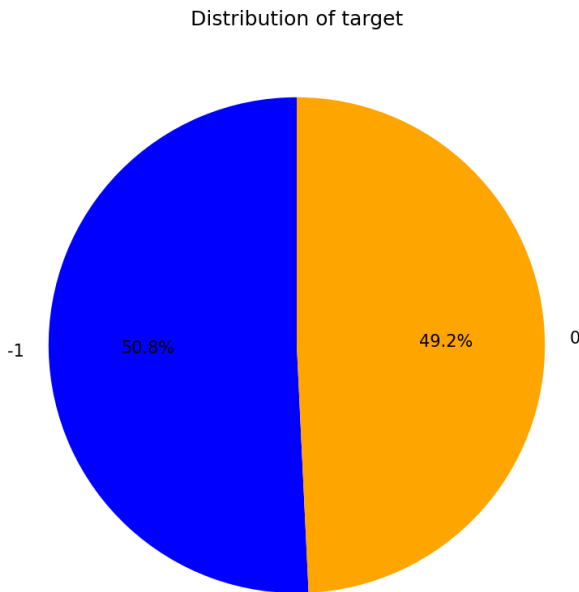


FIGURE 5. A pie chart showing target distribution

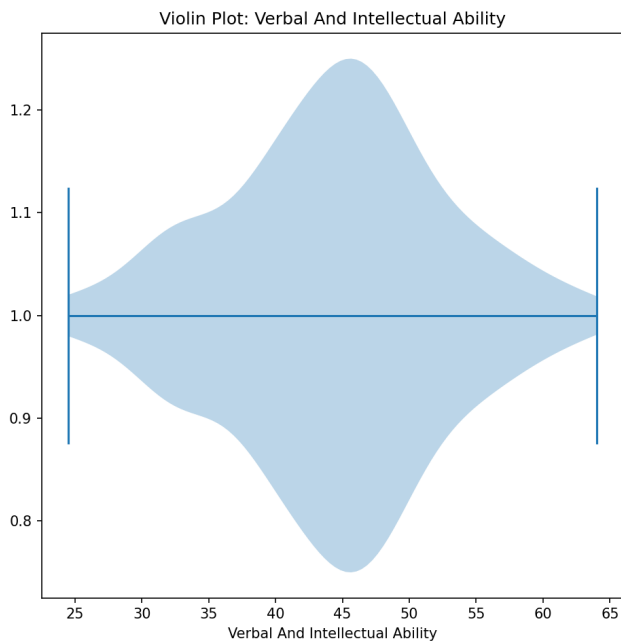


FIGURE 6. Violin plots to show variance in memory and verbal and intellectual ability

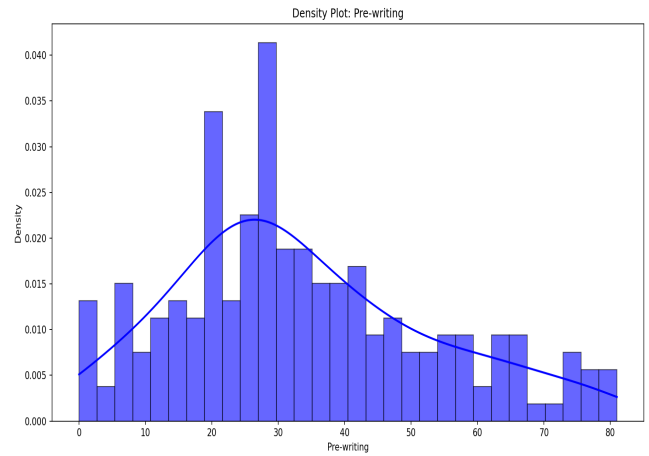


FIGURE 7. A density plot for pre-writing

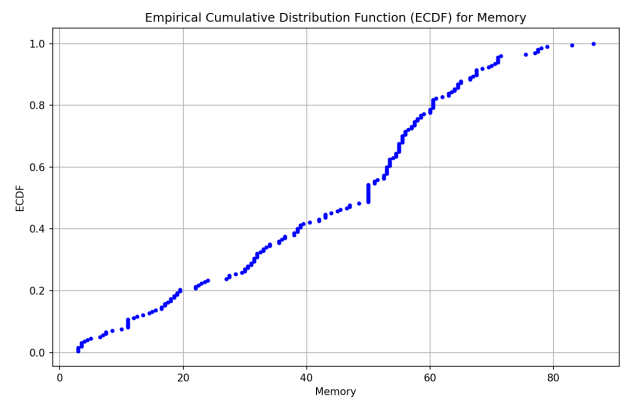


FIGURE 8. An empirical cumulative Distribution function (ECDF) for memory

grapple with learning disabilities.

In figure 8, data is ordered from least to greatest to see the whole feature as if is distributed across the data set. The ECDF shows the portion of individuals score below or at a certain memory level. A steep increase suggests that multiple individuals share the same memory score. From the ECDF it is observed that a large number of individuals had very low memory scores which could be a sign of learning disabilities.

In figure 9 the rug plot illustrates individual data points represented by vertical lines along one axis to visualize data. The dense vertical lines indicate the areas with more frequent occurrences of verbalization. The sparse areas indicate the less common occurrences.

A. DETECTING OUTLIERS

Box plots and scatter plots for each variable were plotted to visualize the spread of the data and detect outliers. The outliers detected were neither removed nor adjusted because in medical-related research such as this, extreme values might indicate uncommon but important conditions. Any form of tampering with them could bias the dataset and make the

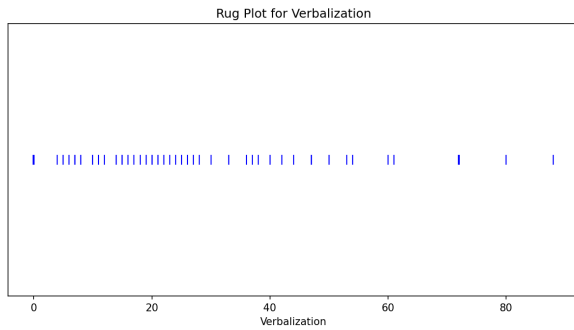


FIGURE 9. A rug plot visualization for Verbalization

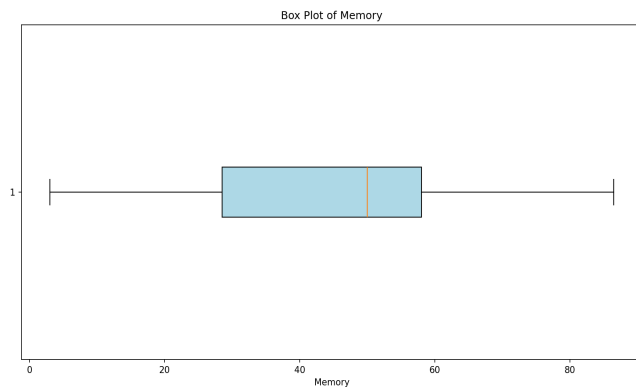


FIGURE 10. A Box plot to show data spread of memory

model less accurate and reliable.

In figure 10 the box plot shows how data is spread in the memory feature. A box plot is used to understand the data structure and spot potential outliers in the features. They consist of medians and whiskers plus boxes that are used to understand the spread of data. For the distribution, the median is slightly above the center of the box. The whiskers are symmetrically spread out to indicate that data is spread evenly without extreme outliers.

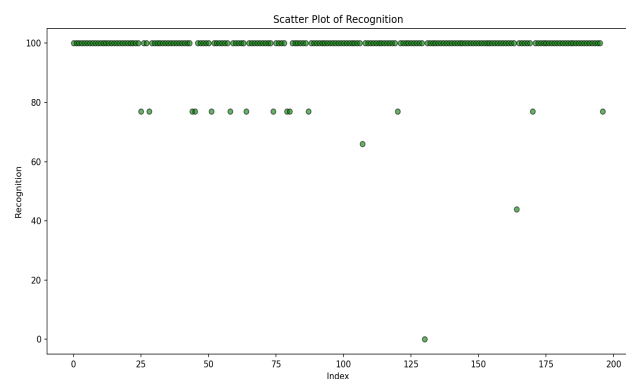


FIGURE 11. A Scatter plot visualizing data spread in the recognition feature

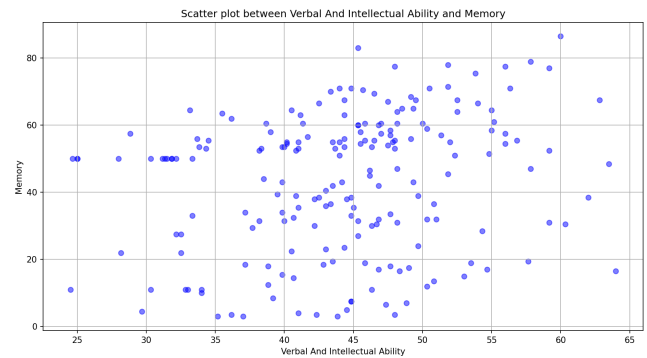


FIGURE 12. A scatter plot to show the correlation between verbal and intellectual ability and memory

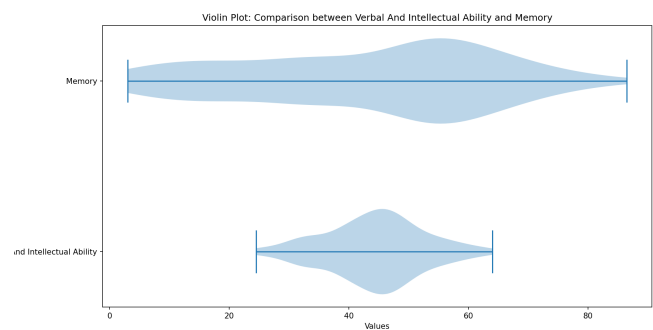


FIGURE 13. A violin plot for memory and verbal and intellectual ability

Figure 11 shows an alternative to box plots. Scatter plots can be used to show the distribution of data and spot potential outliers too. The points clustered together are where normal data lies while those spaced indicate outliers. For the distribution showing recognition, most individuals had the ability to recognize data and scored highly. Some few extreme cases could not. Those could be individuals with learning disorders.

B. BIVARIATE ANALYSIS

The next strategy used was bivariate analysis to understand the relationship between two variables to find connections, correlations, and dependencies. The two variables inspected here were Verbal and intellectual ability and memory.

In figure 12 scatter plot are used to examine the relationship between variables, in this case they are verbal and intellectual ability and memory. Because the plots are moving upwards, there is a positive correlation between the two features. This means that if an individual suffers with verbal and intellectual ability, then they most probably suffer with deficit memory.

In figure 13 the violin plot in figure illustrates the variance between memory and verbal and intellectual ability and memory. The violin plot for memory is rightly skewed indicating that most individuals scored highly in the test. However, there are those that did not. From the violin plot,

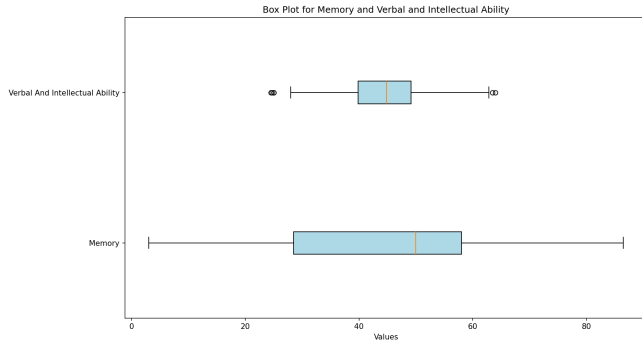


FIGURE 14. A box plot to show memory and verbal and intellectual ability and memory

it is observed that those that the region covered by those that suffer with memory deficit spans that of individuals that have both high and low scores when tested on verbal and intellectual ability. This implies that some individuals although excellent at communicating and reasoning may still suffer with memory deficit and may have learning disabilities. It also implies that individuals poor at communicating and reasoning can also grapple with memory deficit. Still, these too could have learning disabilities.

Figure 14 shows how data is spread in the memory and cognitive flexibility features. In the box plot for memory the median is slightly above the centre of the box implying that most individual sin the study experience memory deficit. The region covered by these individuals spans that of individuals that have both high and low scores when tested on verbal and intellectual ability. This implies that some individuals although excellent at communicating and reasoning may still suffer with memory deficit and may have learning disabilities. It also implies that individuals poor at communicating and reasoning can also grapple with memory deficit. Still, these too could have learning disabilities. For both the visualization, their whiskers are spread out evenly indicates a normal distribution without outliers

From the hexbin plot in figure 15, the count in each bin ranges from 0 to 5 to show the areas where the two features are more likely to occur together. The bins with higher counts indicate higher concentration of datapoint swchich implies a more frequent occurrence of verbal and intellectual ability and memory together in an individual grappling with learning disabilities.

In figure 16 the joint plot is used to visualize data and identify relationships two features. For individuals with learning disabilities, the curve shows instances when one may have a learning disability and at the same time either suffers with poor verbalisation or not. The curve from -50 to 50 indicates instances when individuals with learning disabilities are affected by failure to use language in social communication context. From 50 to 125, these individuals have learning disabilities but are not affected by poor verbalisation. For individuals without learning disabilities (0), the curve shows

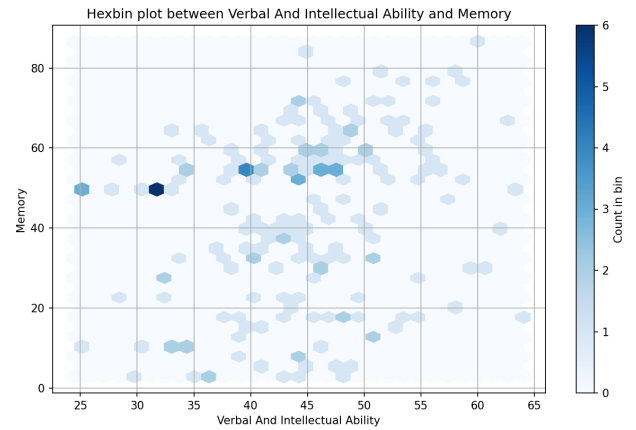


FIGURE 15. A visualization of a hexbin plot between verbal and intellectual ability and memory.

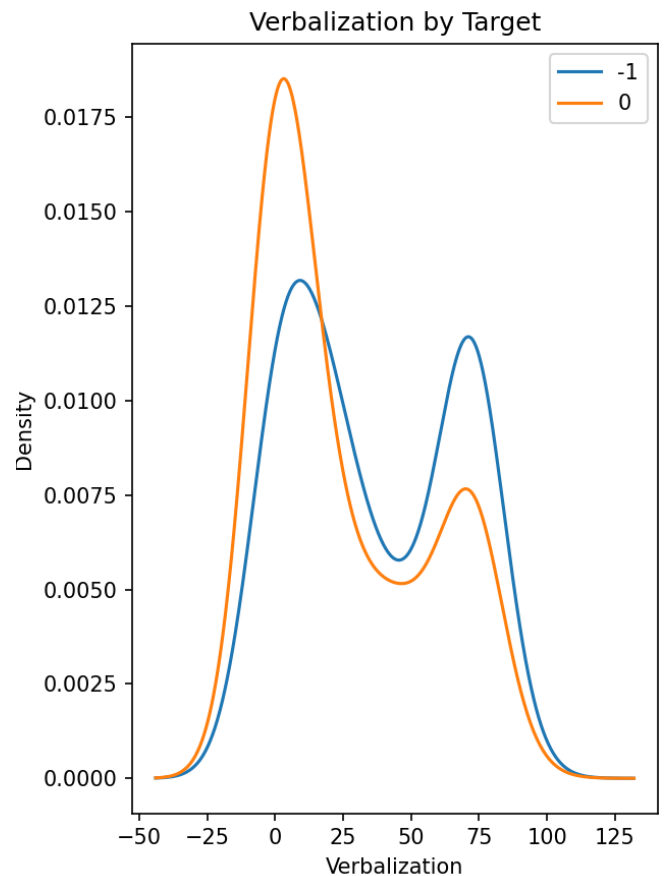


FIGURE 16. A Joint plot for target and verbalization

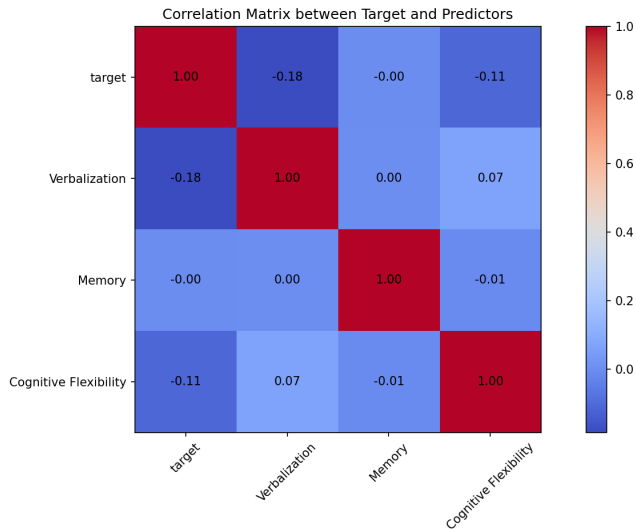


FIGURE 17. A correlation matrix to show the relationship between target and predictors

instances when one may not have a learning disability and at the same time either suffers with poor verbalisation or not. The curve from -50 to 50 indicates instances when individuals without learning disabilities are affected by failure to use language in social communication context. From 50 to 125, these are not affected by poor verbalisation.

C. MULTIVARIATE ANALYSIS

The last strategy used was multivariate analysis. Here, more than two variables were examined. Target, Verbalization, Memory, and Cognitive Flexibility. A correlation matrix was plotted to visualize all the variables at once to see how strongly they relate. The goal was to see which features are important in the detection of learning disabilities. The features with a high positive correlation can be used to train the model on what to learn and those with a high negative correlation can be used to train a model on what not to learn.

Figure 17 multivariate analysis is done using a correlation matrix. Values close to +1 indicate strong positive correlation, while those closer to -1 indicate a strong negative correlation while 0 indicates suggests no linear correlation. The darker colors signify strong correlation, while light colors represent weaker correlations. Positive correlation variable move in same directions. As one increases, the other also increases. Negative correlation variable move in opposite directions. An increase in one variable is associated with a decrease in the other. This correlation matrix aims to show the relationship between a learning disability (target) i.e; whether one has one or not and the signs used to predict its occurrence. It is observed that verbalization has a weak negative correlation with the target (-0.18), memory shows no correlation with the target (-0.00), and cognitive flexibility has a small negative correlation with the target. This suggests

3D Scatter Plot between Verbalization, Memory, and Cognitive Flexibility

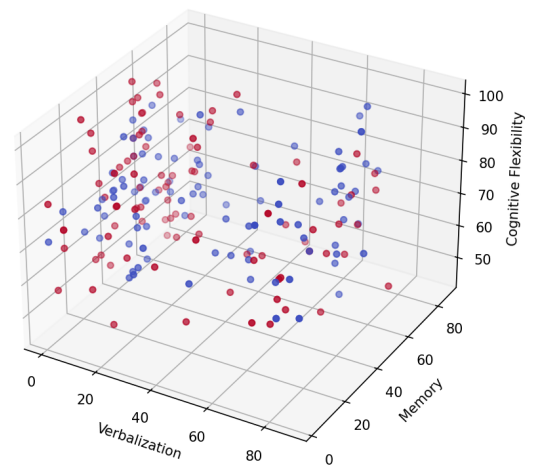


FIGURE 18. A 3D scatter plot between visualization, memory and cognitive flexibility

Radar Chart for Target vs Predictors

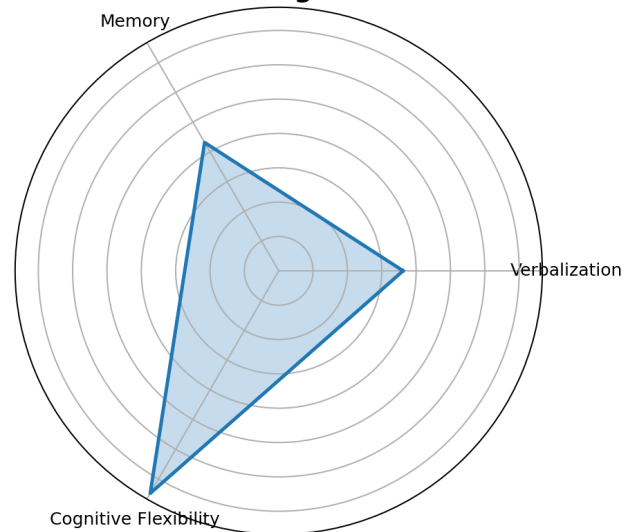


FIGURE 19. A radar chart for target vs predictors

that the predictors might not be strong indicators for the target.

In figure 18 the 3D scatter plots is used to plot data points on three axes in the attempt to show the relationship between three variables. From the image, it is observed that there is no clear relationship between the three variables.

From figure 19, it is observed that memory and cognitive flexibility are closer to the plot while verbalization is further away. This implies that verbalization has a more distinct role in influencing the target. It could influence it negatively or positively.

The multi histogram plots in figure 20 are used to show

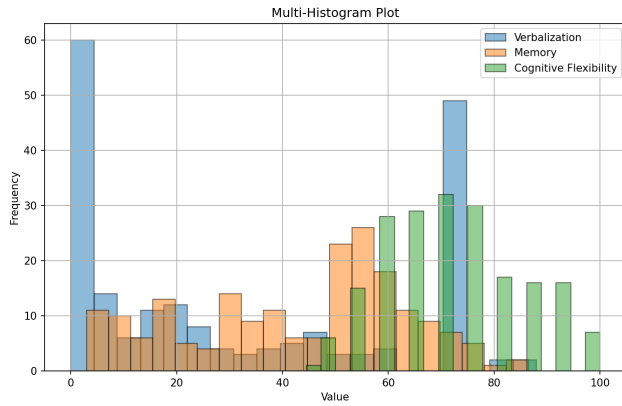


FIGURE 20. A multi histogram plot for verbalization, memory and cognitive flexibility

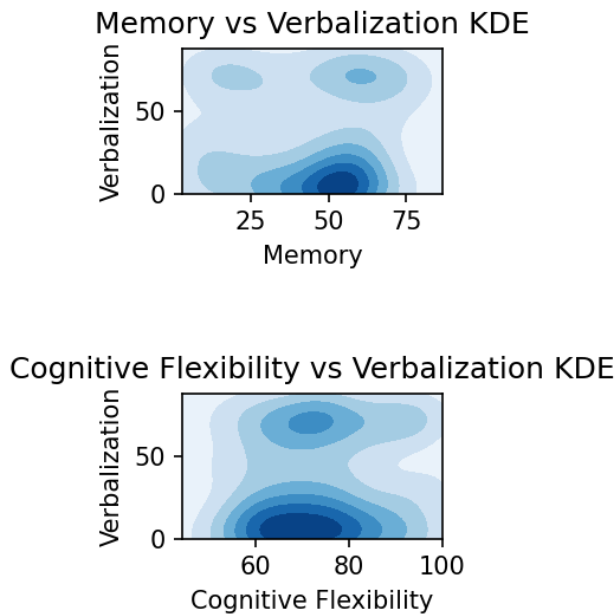


FIGURE 21. Pairwise KDE plots for verbalization, cognitive flexibility and memory

multiple features within the same plot, making it easier to compare them clearly. It is observed that memory deficit is the most common occurrence, followed by cognitive flexibility and lastly verbalisation

In figure 23 the pairwise KDE plots are used to visualize data and identify relationships between one variable against others. The darker colors indicate the instances when the two features occur together in their highest density. Cognitive flexibility and Verbalization occur more frequently as compared to memory deficit with verbalization.

From the visualizations and analyses, it is observed that memory deficits were common with no clear relationship to verbal and intellectual ability. This implies that mem-

ory deficits are common in both high and low performing individuals indicative of potential learning disabilities. The correlation matrix revealed weak correlations between the target (learning disability) and the predictors suggesting that a combination of features may be necessary to predict learning disabilities accurately. Despite the weaker correlation of verbalization with the target, it might have a significant role in distinguishing individuals with learning disabilities, as suggested by the radar chart and hexbin plot. The 3D scatter plot and pairwise KDE plots helped visualize relationships between multiple features, though no clear patterns emerged, implying the complexity of learning disabilities and the need for further analysis and more complex models.

VIII. MODELS

After understanding the dataset, and its features, models were built to predict learning disabilities in children based on observed symptoms. Different learning disabilities manifest with distinct features, some sharing overlapping symptoms. To address this, retrieval augmented generation with cache augmented generation and fine-tuned DeepSeek 8B, a transformer based model with 8 billion parameters were used. This approach ensures the model retrieves relevant medical literature while engaging users in an interactive diagnostic process.

A. MODEL 1: NON-FINETUNED RAG WITH CACHE AUGMENTED GENERATION

The first model was built to rely on cache retrieval augmented generation without fine-tuning. When a user inputs a symptom-based query, to the model which is then passed to the embedding model to convert it to vector representations which are passed to the query engine by the model. The query engine uses this to query the vector store and retrieve documents similar to the query. The documents retrieved from the first three queries are stored in cache which the model uses to ask a follow up questions to identify the learning disability present. The model asks a maximum of fifteen well structured queries imputed by the user which are stored by cache and at the end, a well structured report containing the follow up questions asked by the model and user responses that a psychiatrist can base off of to make diagnosis. The model also provides a step by step rundown of it's reasoning process to give the psychiatrists an insight on its thought process to avoid black boxing and promote trust. Figure 22 provides a visualization of the mode's architecture. Figure 22 shows the architecture of the model, first the user makes an embedding model that converts documents into numerical representations, a vector storage to retrieve and store documents based on user input. It has a query engine to fetch the top 3 most relevant documents, Cache augmented generation to store retrieved context for efficient use and a model that is not finetuned to structure responses and ask follow-up questions

To implement the model, an environment is set up. Here all the necessary dependencies are installed and libraries to

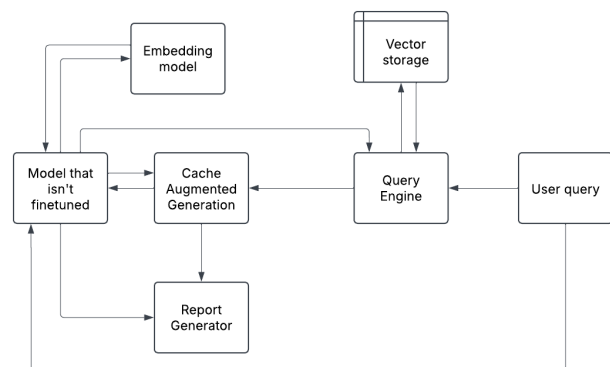


FIGURE 22. A diagram showing the architecture of the model that was not finetuned

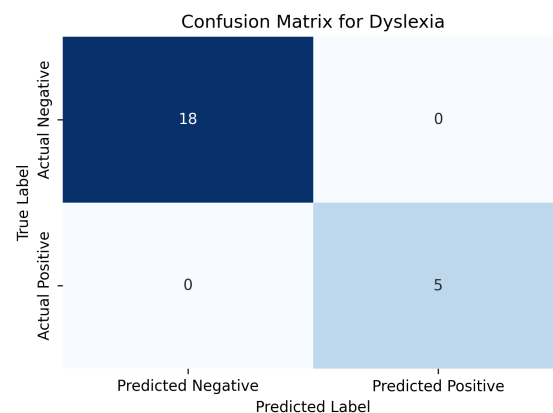


FIGURE 24. A confusion matrix for Dyslexia showing predictions of Dyslexia by the model without fine-tuning

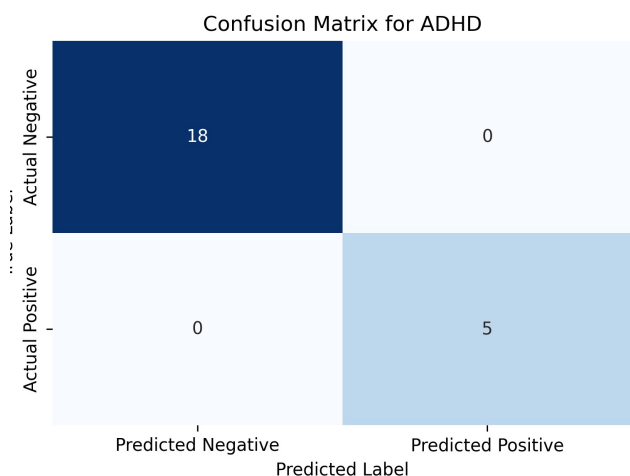


FIGURE 23. A confusion matrix for ADHD showing predictions of ADHD by the model without fine-tuning

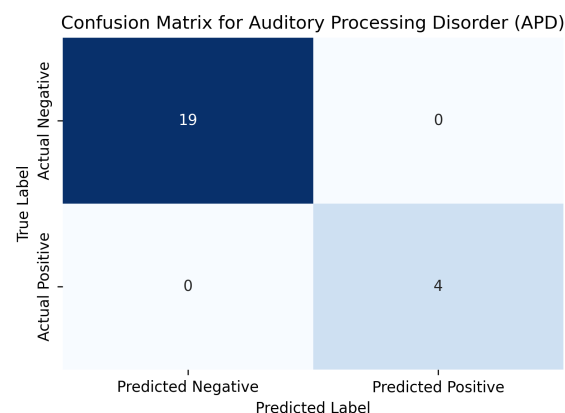


FIGURE 25. A confusion matrix for APD showing predictions of APD by the model without fine-tuning

be used are imported. The next step is to set up the RAG pipeline that will be used to retrieve the context from the large external corpus made available in the vector storage. First, an embedding model is loaded. The external corpus is then loaded and converted to numerical figures by the embedding model. The corpus is then stored in a vector storage. Next, the query engine and retriever are set up so that whenever a user enters a symptom, context relevant documents are retrieved from the vector store. The pipeline is then tested by asking a question and the documents retrieved are checked to see if they are contextually accurate. After verifying that everything is as it should be, the model is loaded through a process called inference. A prompt is generated to act as a guide for the model on how to structure its responses. This time the prompt includes the context provided by the documents retrieved by the model. The model is then queried with multiple user symptom-based statements to evaluate its accuracy.

In figure 23, 24, 25, 26, and 27 the model accurately pre-

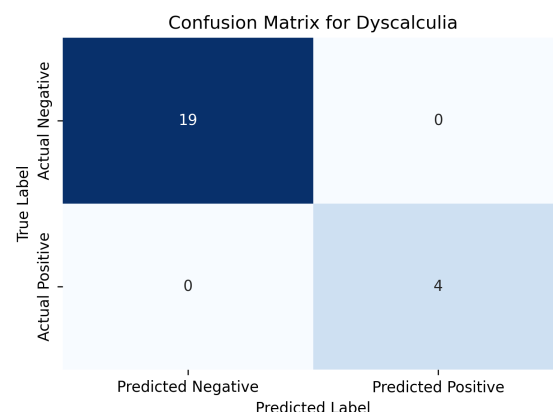


FIGURE 26. A confusion matrix for dyscalculia showing predictions of dyscalculia by the model without fine-tuning

Algorithm 1 Development of the Non-Finetuned RAG Model

```

1: Initialize: Vector DB =  $\emptyset$ , Cache =  $\emptyset$ 
2: Preprocessing:
3: Load dataset
4: Split text into chunks of size  $C$  with overlap  $O$ 
5: Embed documents  $D = \{d_1, d_2, \dots, d_n\}$  using embedding model  $E$ 
6: Store embeddings in Vector DB
7: Query Processing:
8: for  $i = 1$  to MaxQuestions do
9: Receive user query  $q_i$ 
10: Embed query:  $E(q_i)$ 
11: Retrieve top- $k$  documents:  $R_i = \text{TopK}(E(q_i), \text{Vector DB}, k)$ 
12: Construct prompt  $P_i = \text{Template}(q_i, R_i, \text{Cache})$ 
13: Generate response using LLM:  $r_i = \text{LLM}(P_i)$ 
14: if  $|\text{Cache}| < \text{MaxCacheSize}$  then
15: Cache = Cache  $\cup \{(q_i, r_i, R_i)\}$ 
16: end if
17: if Diagnosis complete then
18: break
19: end if
20: end for
21: Return: Final report  $F = \text{GenerateReport}(\{(q_1, r_1), \dots, (q_n, r_n)\}) = 0$ 

```

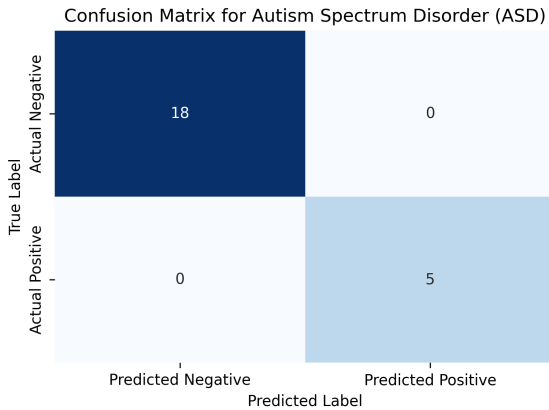


FIGURE 27. A confusion matrix for ASD showing predictions of ASD by the model without fine-tuning

dicted all present cases of ADHD, dyslexia, APD, dyslexia, and ASD. It did not predict any false negatives or false positives. This shows that the model generalises well across different cases and effectively uses contextual information from the user response and information provided by the RAG pipeline. Absence of miscalculations suggests strong pattern recognition capabilities and good decision making to differentiate between different learning disabilities.

B. MODEL 2: FINETUNED RAG WITH CACHE AUGMENTED GENERATION

The second model was built running on *DeepSeek-R1-Distill-Llama-8B*. A prompt is defined which will be used to help the model understand its task and guide it on how to answer user queries. The model is then tested through a process called inference; a query is passed to the model formatted in the structure of the prompt. When responding, the model uses the prompt as a template to follow when answering the question. When satisfied with the response, training can begin. An even better training prompt is generated as a guide for how the trained model will respond to responses once inferred. The data set is then loaded and formatted to fit the training prompt. Because even when distilled, the model is still large, quantization using low-rank adaptation techniques is not to optimize the model for training. Then the few parameters chosen for fine-tuning are chosen. Training arguments are set up, and the model is trained. The trained model is now used for developing the RAG pipeline described in the previous section. Figure 28 shows the architecture of the model. It is similar to the model that is not finetuned except that it uses a finetuned model.

Algorithm 2 Development of the Finetuned RAG Model

```

1: Initialize: Vector DB =  $\emptyset$ , Cache =  $\emptyset$ 
2: Model Finetuning:
3: Load dataset  $D$ 
4: Define LoRA parameters
5: Set up training arguments  $A$ 
6: Finetune model  $M$  using  $(D, A)$ 
7: Save finetuned model  $M'$ 
8: RAG Setup:
9: Initialize embedding model  $E$ 
10: Convert dataset to vector representations  $V = \{E(d_1), E(d_2), \dots, E(d_n)\}$ 
11: Store embeddings in Vector DB
12: Define retrieval parameter  $k$ 
13: Query Processing:
14: for  $i = 1$  to MaxQuestions do
15: Receive user query  $q_i$ 
16: Embed query:  $E(q_i)$ 
17: Retrieve top- $k$  documents:  $R_i = \text{TopK}(E(q_i), \text{Vector DB}, k)$ 
18: Construct prompt  $P_i = \text{Template}(q_i, R_i, \text{Cache})$ 
19: Generate tokens:  $o_i = M'.\text{Generate}(\text{Tokenizer}(P_i))$ 
20: Decode response:  $r_i = \text{Tokenizer}.\text{Decode}(o_i)$ 
21: if  $|\text{Cache}| < \text{MaxCacheSize}$  then
22: Cache = Cache  $\cup \{(q_i, r_i, R_i)\}$ 
23: end if
24: if Diagnosis complete then
25: break
26: end if
27: end for
28: Return: Final report  $F = \text{GenerateReport}(\{(q_1, r_1), \dots, (q_n, r_n)\}) = 0$ 

```

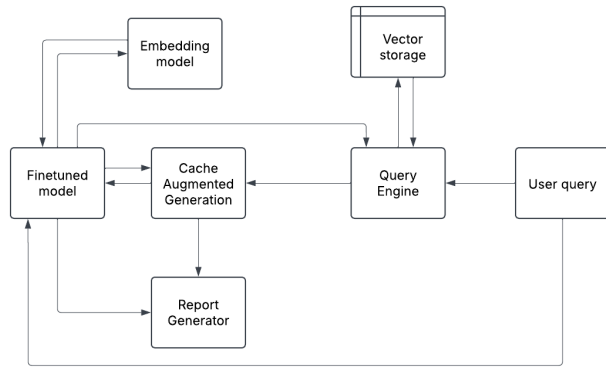


FIGURE 28. A diagram showing the architecture of the fine-tuned model

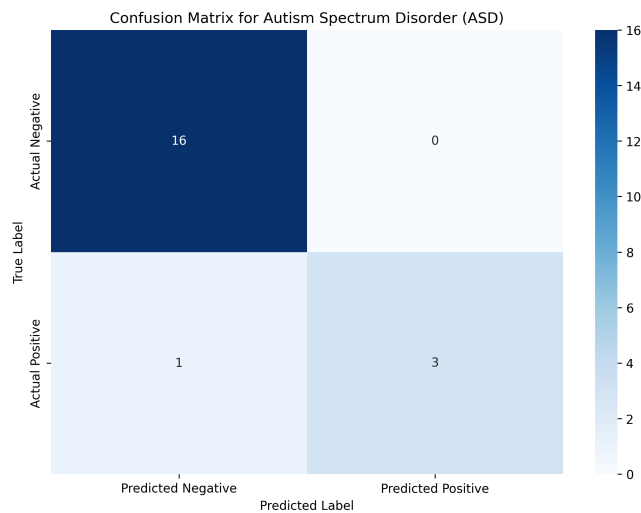


FIGURE 29. A confusion matrix for ASD showing predictions of ASD by the finetuned model

In figure29, 30, 31 and 41 the model was also able to accurately predict most present cases of dyslexia, ASD, dyscalculia, and ADHD with no false negative results or false positive. The model was almost accurate at identifying the learning disabilities.

In figure40 the model was also able to accurately predict most present cases of APD with some false positive results but no false negative. The model was quite accurate at identifying the learning disability.

C. MODEL EVALUATION

Table 2 shows the accuracy, performance metric precision, recall, F1 score and specificity of the model that was dependant of a fine-tuned model alongside cache and retrieval augmented generation. The accuracy shows that the model performed exceptionally well in identifying instances of learning disabilities. The high precision indicates a low false positive rate, meaning that when the model predicted a learning

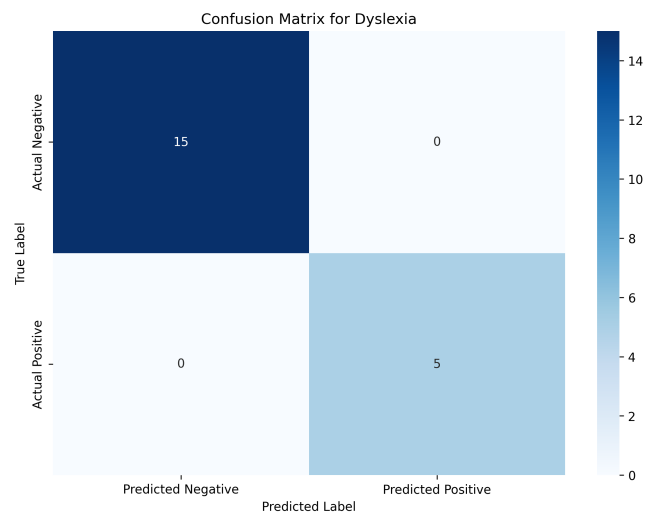


FIGURE 30. A confusion matrix for Dyslexia showing predictions of dyslexia by the finetuned model

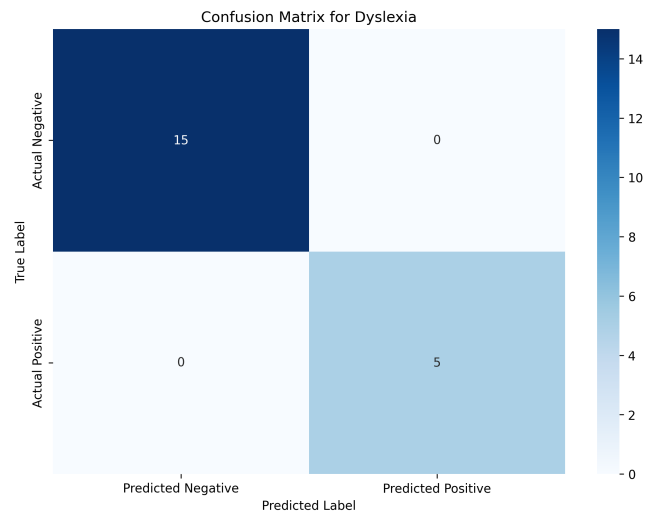


FIGURE 31. A confusion matrix for dyscalculia showing predictions of dyscalculia by the fine-tuned model

TABLE 2. Performance metrics of the model dependent on a fine-tuned model with cache and retrieval augmented generation

Performance Metric	Score Value
Accuracy	0.9500
Precision	0.9600
Recall/ Sensitivity	0.9500
F1 Score	0.9492
Specificity	0.9875
Rouge-1	0.2317
Rouge-2	0.0554
Rouge-L	0.0554
Toxicity	0.002184
Bias	0.0

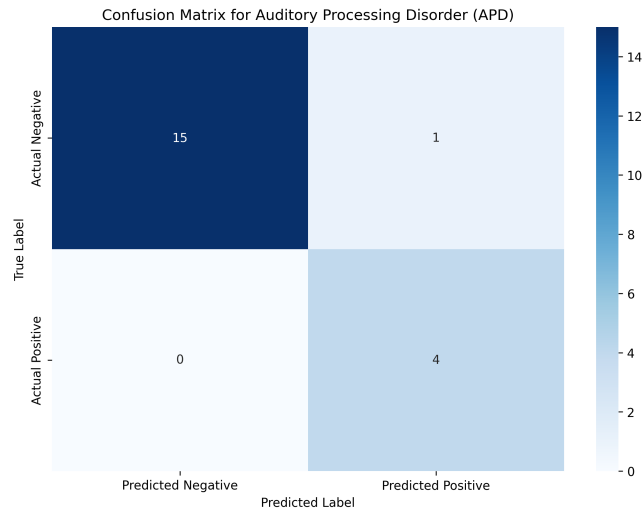


FIGURE 32. A confusion matrix for APD showing predictions of APD by the fine-tuned model

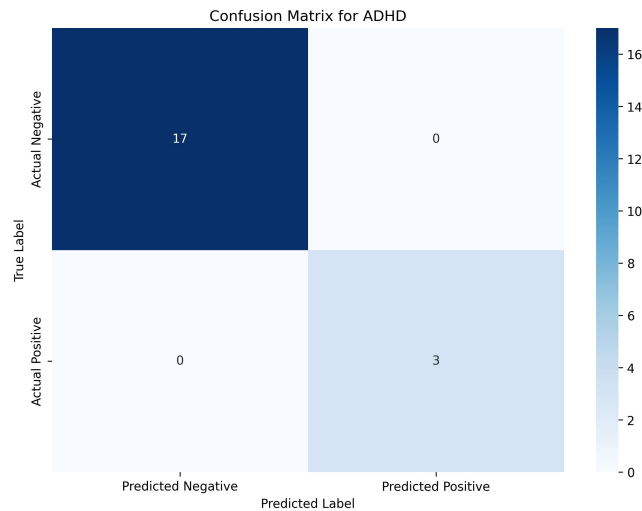


FIGURE 33. A confusion matrix for ADHD showing predictions of ADHD by the fine-tuned model

disability, it was correct most of the time. Sensitivity shows that the model effectively detected learning disabilities. The F1 score demonstrates a good balance between precision and recall, and specificity indicates that the model correctly identified negative cases most of the time.

TABLE 3. Toxicity scores per disorder of the model dependent on a fine-tuned model with cache and retrieval augmented generation

Disorder	Average toxicity score
ADHD	0.00127
Dyslexia	0.00474
Dyscalculia	0.00128
ASD	0.00211
APD	0.00092

TABLE 4. Performance metric scores per disorder of the model that was fine-tuned

Disorder	Accuracy	Precision	F1 Score	Specificity
ADHD	1.00	1.00	1.00	1.00
Dyslexia	1.00	1.00	1.00	1.00
Dyscalculia	1.00	1.00	1.00	1.00
ASD	0.95	1.00	0.86	1.00
APD	0.95	0.80	0.89	0.94

TABLE 5. Performance metric scores per disorder of the model that was not fine-tuned

Disorder	Accuracy	Precision	F1 Score	Specificity
ADHD	0.91	0.80	0.80	0.94
Dyslexia	0.91	0.80	0.80	0.94
Dyscalculia	1.00	1.00	1.00	1.00
ASD	1.00	1.00	1.00	1.00
APD	1.00	1.00	1.00	1.00

Table 3 shows the toxicity scores per disorder for the model that was finetuned. The highest toxicity score belonged to Dyslexia, while the lowest was ASD, followed by ASD. ADHD and dyscalculia averaged at almost the same score but dyscalculia was slightly higher. The toxicity scores were low as they were all below 0.01.

Tables 4 and 5 show the accuracy, precision, f1 score, and specificity of the models per disorder. Accuracy shows the ability of the model to correctly identify the disorders, precision shows the false positive rate (cases where the model identifies a positive instead of a negative occurrence of the disorder) Sensitivity shows the ability of the model to identify the learning disability and the f1 score demonstrates the balance between precision, recall and sensitivity.

Tables 6 and 7 show the ROUGE scores per disorder of the models. The rouge scores evaluate the text generation tasks like summarizaion and machine translation. The rouge-1 scores measure the overlap between the generated text and the reference text from the dataset. The ROUGE-2 scores show the overlap of two-word sequences between the generated text and reference text, and the ROUGE-L evaluates how

TABLE 6. Average ROUGE scores per disorder of the model that was fine-tuned

Disorder	ROUGE-1 score	ROUGE-2 Score	ROUGE-L score
ADHD	0.2209	0.0747	0.1870
Dyslexia	0.2237	0.0200	0.1759
Dyscalculia	0.2261	0.0255	0.1767
ASD	0.2565	0.1075	0.2565
APD	0.2406	0.0824	0.2103

TABLE 7. Average ROUGE scores per disorder of the model that was not fine-tuned

Disorder	ROUGE-1 score	ROUGE-2 Score	ROUGE-L score
ADHD	0.1319	0.0255	0.1275
Dyslexia	0.2122	0.0476	0.1911
Dyscalculia	0.1878	0.0355	0.1792
ASD	0.2063	0.0629	0.1950
APD	0.1921	0.0499	0.1855

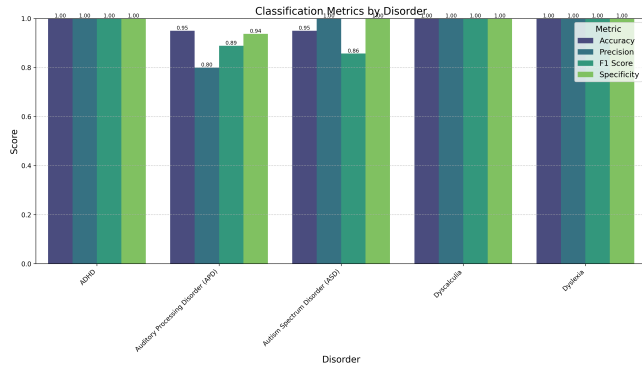


FIGURE 34. A disorder comparison showing performance metrics of the model that was fine-tuned

TABLE 8. Performance metrics of the model that was not fine-tuned

Performance Metric	Score Value
Accuracy	0.9565
Precision	0.9638
Recall/ Sensitivity	0.9565
F1 Score	0.9561
Specificity	0.9889
ROUGE-1	0.1890
ROUGE-2	0.0461
ROUGE-L	0.1780
Toxicity	0.0008924
Bias	0.0

well the generated text maintains the structure and fluency of the reference text.

Figure 34 shows the accuracy, precision, recall, F1 score and specificity of each learning disability using the disorder label field in the dataset as the ground truth value and the predicted value as the disorder the model predicts after it is sent a user response from the dataset. The model did generally well at predicting all disabilities but a few.

Table 8 shows the accuracy, performance metric precision, recall, F1 score and specificity of the model was not fine-tuned. The accuracy shows that the model performed exceptionally well in identifying instances of learning disabilities. The high precision indicates a low false positive rate, meaning that when the model predicted a learning disability, it was correct most of the time. Sensitivity shows that the model effectively detected learning disabilities. The F1 score demonstrates a good balance between precision and recall, and specificity indicates that the model correctly identified negative cases most of the time.

TABLE 9. Average toxicity scores per disorder of the model that was not fine-tuned

Disorder	Average toxicity score
ADHD	0.003033
Dyslexia	0.000805
Dyscalculia	0.000795
ASD	0.000930
APD	0.000805

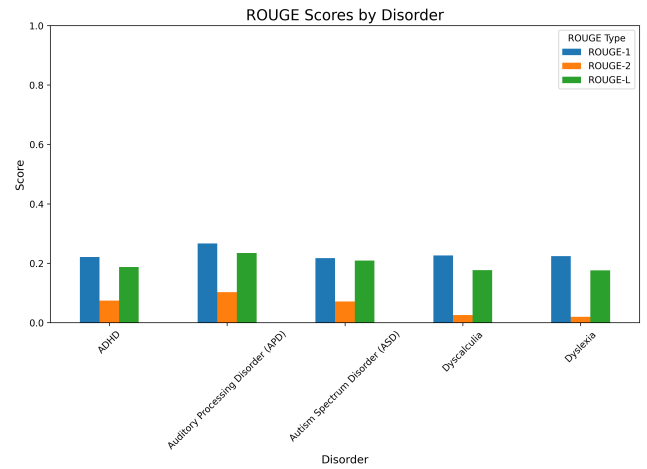


FIGURE 35. ROUGE scores per disorder of the model that was fine-tuned

In table 9 the highest toxicity score belonged to ADHD. APD and Dyslexia averaged at the same score, the second highest was ASD followed by dyscalculia. Although the score for ADHD is quite high, the rest of the disorders average at almost the same score, meaning the model is able to maintain a balance between the other four disorders. This table indicates that there was some toxicity present in the responses generated by the model but since the scores are below 0.1, the overall toxicity is quite low.

The graph in figure 35 shows the rouge scores of the model. The x-axis shows the disorder being evaluated and the Y-axis shows the rouge scores. Rouge-1 (blue) measures the unigram overlap, it is the highest amongst all scores suggesting a good single word similarity. Rouge-2 (orange) measures overlap of two word sequences between the generated text and the reference text, it is significantly lower than the rouge-1 score suggesting that the chatbot struggles with generating multiple word sequences that match the reference text exactly. Rouge-L measures the longest common subsequence it is slightly lower than rouge-1 but higher than rouge-2 which in turn suggests structural similarity. The chatbot performs slightly better on some disorders like APD and ASD where rouge-1 and rouge-2 are higher. The figure 36 shows a bar graph of toxicity scores per disorder. Dyslexia had the highest toxicity score while APD had the lowest. ADHD and Dyscalculia averaged at the same levels with ASD having the second highest toxicity level. Since the overall toxicity is below 0.01, the responses of the model were barely toxic.

The figure in 37 shows the accuracy, precision, recall, F1 score and specificity of each learning disability using the disorder label field in the dataset as the ground truth value and the predicted value as the disorder the model predicts after it is sent a user response from the dataset. This was the model that was not fine-tuned. Although both models seem quite accurate, the fine tuned model struggled immensely to follow prompts and adapt to conversational dialogues between the user and the model itself. The non-fine-tuned model required

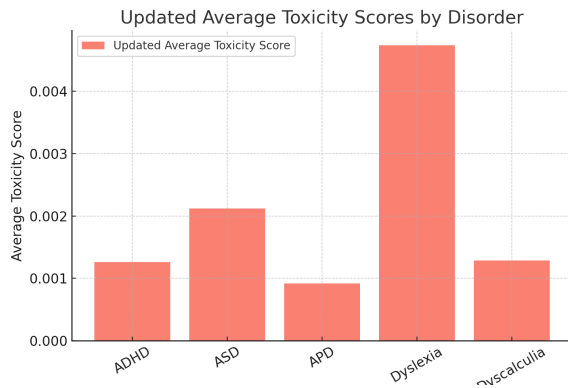


FIGURE 36. Toxicity scores per disorder of the model that was fine-tuned

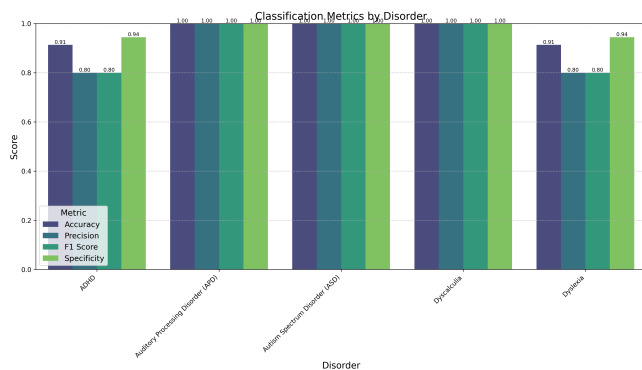


FIGURE 37. A disorder comparison showing performance metrics of the model that was not fine-tuned

a bit of tweaking but definitely generalised better to the concept of having conversations with the user. Instead of asking follow up questions, sometimes the model would plain out diagnose the user. While the non-fine-tuned model took it's time making sense of the context and user query to ask contextually accurate questions. The non-fine-tuned model did better with using context to respond to user queries.

The graph in figure 38 shows the rouge scores of the model. The x- axis shows the disorder being evaluated and the Y axis the rouge scores. Rouge- 1 (blue) measures the unigram overlap, it is the highest amongst all scores suggesting a good single word similarity. Rouge-2(orange) measures overlap of two word sequences between the generated text and the reference text, it is significantly lower than the rouge-1 score suggesting that the chatbot struggles with generating multiple word sequences that match the reference text exactly. Rouge-L measures the longest common subsequence it is slightly lower than rouge-1 but higher than rouge-2 which in turn suggests structural similarity. The chatbot performs slightly better on some disorders like ASD and dyslexia where rouge-1 and rouge-2 are higher.

Overall, in figure 39 ADHD had the highest toxicity score implying that responses concerning ADHD contained

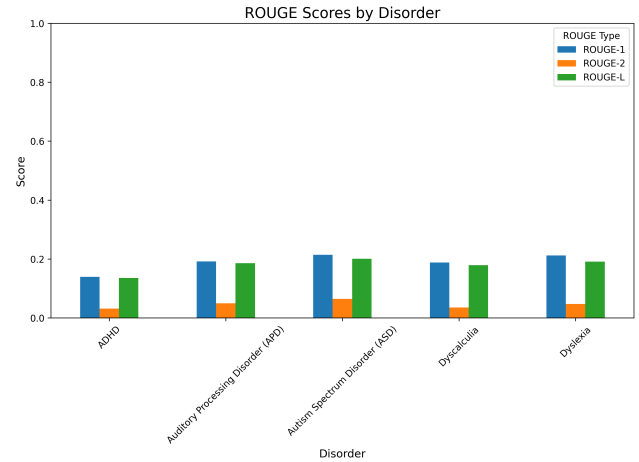


FIGURE 38. A disorder comparison showing ROUGE scores per disorder of the model that was not fine-tuned

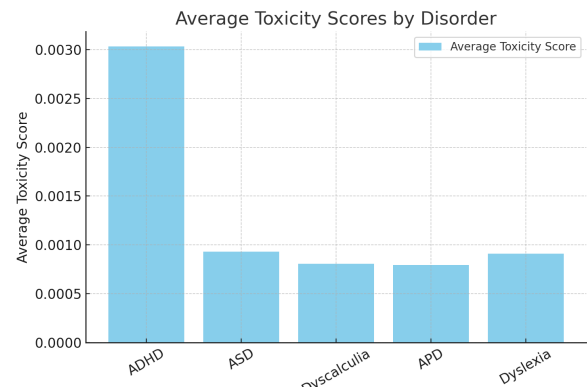


FIGURE 39. A disorder comparison showing toxicity scores per disorder of the model that was not fine-tuned

slightly more toxic language than the other disorders. The other disorders have lower and almost similar toxicity scores. Since all the toxicity scores are below 0.01, the toxicity across all the disorders is minimal.

D. MODEL SELECTION

Based off of the performance metrics results, both models performed exceptionally well during the prediction of the disorders based off of the testing dataset. However, the finetuned model performed better than the non-finetuned model. This could be because of problems related to context generation with the RAG pipeline, or maybe the way in which the data was structured within the corpus. The fine-tuned model had better precision scores for the different disabilities. However, the toxicity scores of the model that was not finetuned averaged almost at the same level with the different disorders except ADHD while those of the model that was finetuned fluctuated from high to low according to the different disorders. This makes the model that isn't finetuned more reliable

than the one that was. Using the ROUGE scores, it can be seen that the model that was finetuned provided more contextually relevant responses that matched some of the reference text in the dataset. This implies that the responses generated by the fine-tuned model were more contextually accurate than those provided by the model that was not finetuned. The optimal choice between the two models that was finetuned based off of the performance metrics. This is because, in a real world setting, if the model is more likely to succeed in user interaction. Additionally, when tested conversationally, the non-finetuned model went ahead and diagnosed users without following the set limit of 15 questions before generating a report for the possible learning disabilities. The chatbot is only meant to be used as a support tool, and never a diagnosis tool. In most regions, self diagnosis is illegal. The aim of the chatbot is to accelerate the the diagnosis process, not take up the job of the psychiatrist entirely. In the case that the chatbot wrongfully diagnoses a child, he or she may be put on medication that they don't even need. The chatbot is just a start, there are multiple steps in diagnosing learning disabilities. Say dyslexia for example, it can be diagnosed a number of ways, having a conversation is just the start. Handwriting analysis can be done to detect learning disabilities. For Auditory processing disorder, certain frequencies of sound can be used to get an accurate diagnosis [35]. Even in ADHD, neural brain imaging can be done to detect signs of the learning disability [36]. An inaccuracy in this preliminary step would subject a child to far more steps in the diagnosis chain that they may not actually need because of a misdiagnosis. For these reasons, the model that was fine-tuned might most likely do better in the real world and provide accurate results.

IX. XAI TECHNIQUES

Explainable AI is used to describe an AI model, its expected impact and potential biases. It helps characterize model accuracy, fairness, transparency and outcomes in AI-powered decision making. Explainable AI is crucial for an organization in building trust and confidence when putting AI models into production. AI explainability also helps an organization adopt a responsible approach to AI development. Attention visualization, LIME for text, and Text SHAP were used to understand the decision making process of the two models created.

A. ATTENTION VISUALIZATION

To understand how the models make decisions by showing which features (words) it pays attention to when generating an output, logits and softmax probabilities were used. These helped to understand how certain the models were about a given disorder described by the user query in the dataset. Because quantized models were used, it was not possible to access attention weights directly. Instead, token importance was analysed by visualizing the top predicted tokens. First, the user queries were tokenized and then fed into the model. The model uses these user queries to predict the possible

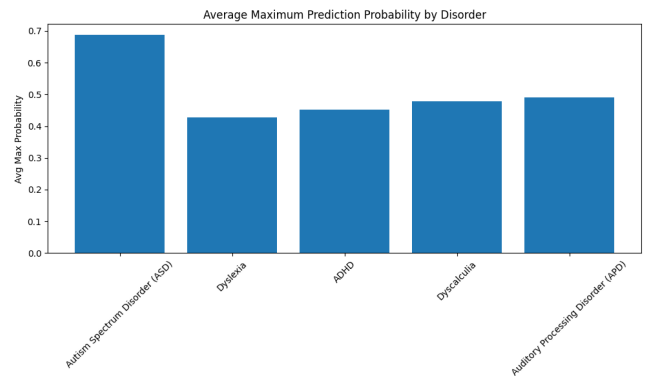


FIGURE 40. The average maximum prediction probability by disorder for the model that was not finetuned

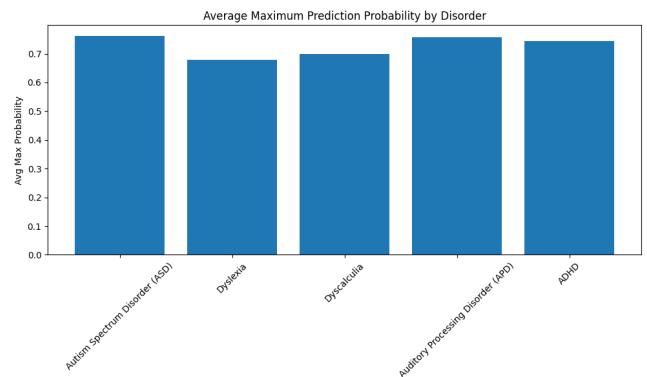


FIGURE 41. The average maximum prediction probability by disorder for the model that was finetuned

learning disability. The raw, unprocessed outputs of a model before it's turned into a probability are called logits. These are extracted and used to understand the model's decision making. Logits from the last token were processed using softmax to generate a probability distribution over the different learning disorders. This conversion helped determine which words contributed most to the final prediction. A bar graph is then plotted to show how confident the model is about each predicted disorder.

The figures 40 and 41 show the average maximum prediction probability by disorder. It measures how confident the fine-tuned model is when predicting different learning disabilities. The Y-axes represent the average confidence level of the model's predictions for each disorder. A value close to 1 means the model is highly confident in its predictions in that disorder. The X-axes show the model's average confidence level for each learning disability. The slight variations in bar height signify that some disorders may have stronger or weaker levels compared to others.

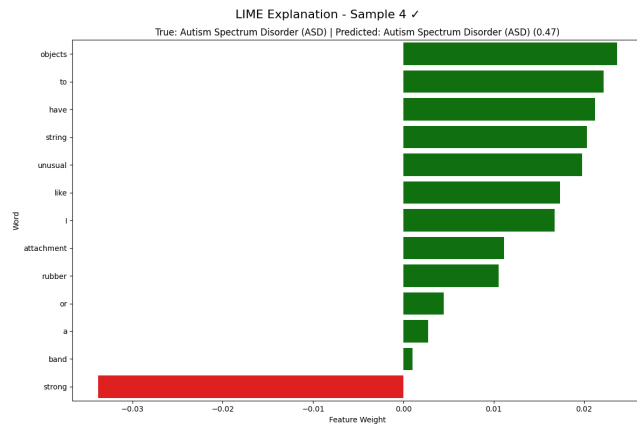


FIGURE 42. LIME (Local Interpretable Model-agnostic Explanations) explanation for the non-finetuned model's prediction of ASD for a specific sample

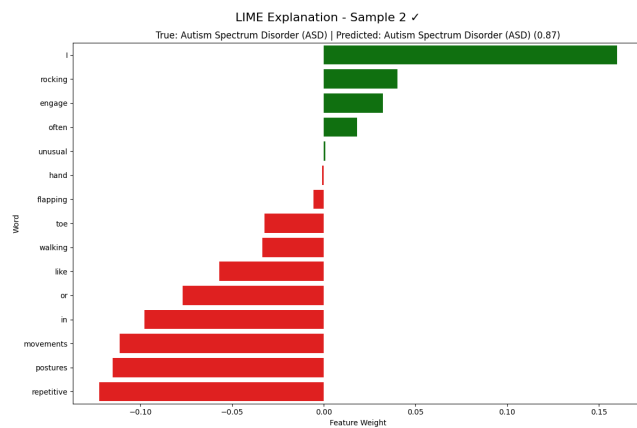


FIGURE 43. LIME (Local Interpretable Model-agnostic Explanations) explanation for the finetuned model's prediction of ASD for a specific sample

B. LIME FOR TEXT

LIME is very useful to help us understand how our model works not only in a common classification problem with the feature presented in tabular format, but it can also interpret how the model works by looking at the text-formatted input. LIME can explain which words in the user query contribute positively or negatively to learning disability prediction. Strong matches to the user query were created by randomly removing words and observing how these changes affected predictions. The model is fitted to these changes to estimate the decision boundary in the local area. Feature weights are also assigned to show how much each word helps or goes against the model prediction. Both correct and incorrect patterns are analysed to identify patterns in the model's decision making process. The overall importance of features is got by combining the model's output and predictions over different user queries. The features that affect correct predictions are measured against those that affect incorrect predictions. The image 43 is a LIME (Local Interpretable Model-agnostic Ex-

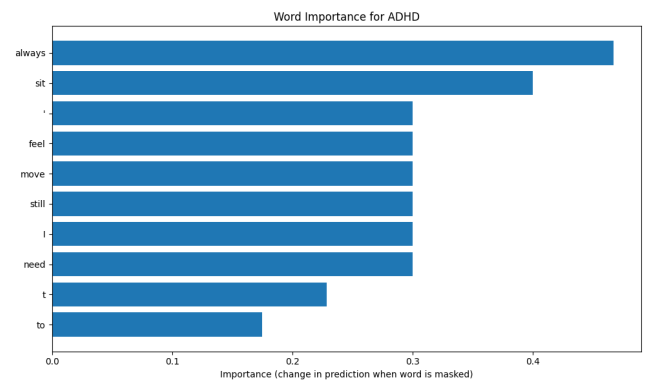


FIGURE 44. A bar chart to show word importance for ADHD classification using SHAP (SHapley Additive exPlanations) for the model that was not finetuned

planations) explanation for the finetuned model's prediction of ASD for a specific sample. The model correctly predicted ASD with a confidence score of 0.87. The horizontal bar chart shows the contribution of individual words from the user query to the model's decision. The green bars represent words that increase the likelihood of ASD prediction while the red bars represent words that decrease the likelihood of the disorder. In figure 43 words like "I," "rocking," "engage," "often," and "unusual" contribute positively to the ASD prediction while words like "repetitive," "postures," or "movements" contributed negatively. Their presence made the model less confident in predicting ASD. In figure 42 words like "objects," "attachment," and "unusual" contribute positively to the ASD prediction while words like "strong" contributed negatively. The model correctly predicted ASD with a confidence score of 0.47.

C. TEXT SHAP

SHAP is a cutting-edge technique for interpreting machine learning models, providing a clear understanding of how each feature (in our case, each word) contributes to the final prediction. Using the SHAP explainer, we can visualize the impact of individual words on the sentiment classification. It assigns importance values to individual words by measuring how each word contributes to a prediction compared to the average prediction. In this case it helps answer the question "Which words in the user query most influenced the model's prediction?"

The figures 44 and 45 are bar charts that visualize the word importance of ADHD and ASD respectively. They represent how much each word influences the prediction of the disorder when removed or masked. The X-axis represents the importance of each word measured by how much the prediction changes when the word is masked. Higher values show stronger influence in the prediction. The y-axis shows the words contributing to the model's prediction. In 44 "always" has the highest importance suggesting that if removed, the ADHD prediction would significantly change. The word

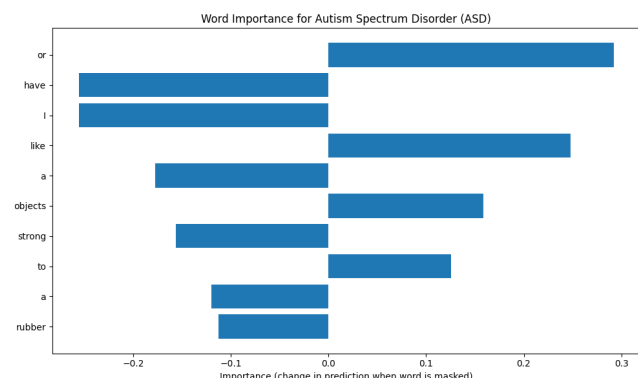


FIGURE 45. A bar chart to show word importance for ASD classification using SHAP (SHapley Additive exPlanations) for the model that was not finetuned

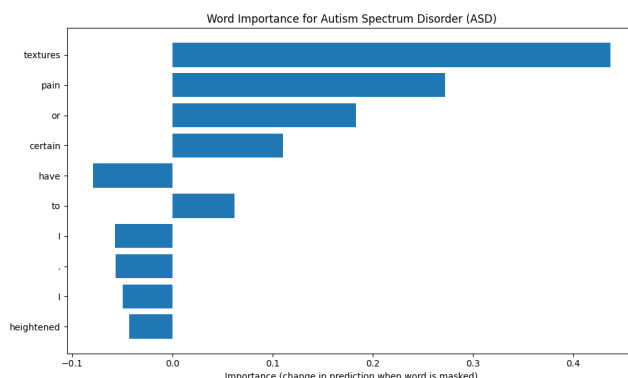


FIGURE 47. A bar chart to show word importance for ASD classification using SHAP (SHapley Additive exPlanations) for the model that was finetuned

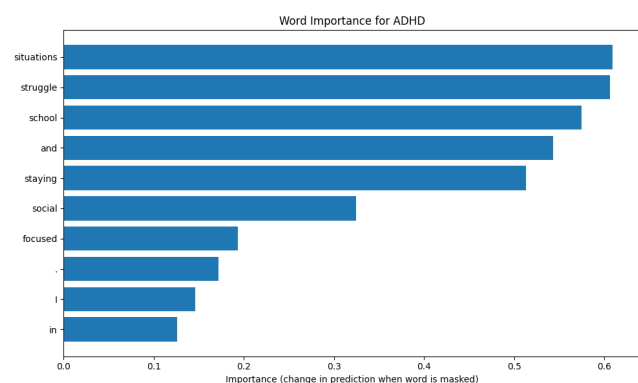


FIGURE 46. A bar chart to show word importance for ADHD classification using SHAP (SHapley Additive exPlanations) for the model that was finetuned

"sit" also plays a crucial role suggesting that discussions around sitting still may be indicative of ADHD symptoms. Words like "feel," "move," and "still" show a relationship with restlessness. The words "always" and "need" may show persistent behavioral patterns. "Move," and "still" suggest that having trouble staying in one place is a key ADHD trait. "t" and "to" could be partial tokenization effects of the large language model. In conclusion, the SHAP analysis shows that words related to movement, impulsivity, and difficulty staying still are strongly related to ADHD prediction. Clinically, the words align with symptoms of hyperactivity and inattention.

Figure 45 and 47 show the importance of specific words in predicting ASD. Words like "textures" and "pain" have higher importance, meaning they significantly impact the prediction of ASD. "Textures" has the highest importance, meaning sensory-related words strongly affect ASD prediction. Words like "have," "to," and "heightened" have smaller importance but they still contribute to the prediction of the disorder. The analysis suggests that features around sensory sensitivity are strong indicators of ASD

The figures 46 and 47 are bar charts that visualize the

word importance of ADHD and ASD respectively. They represent how much each word influences the prediction of the disorder when removed or masked. The X-axis represents the importance of each word measured by how much the prediction changes when the word is masked. Higher values show stronger influence in the prediction. The y-axis shows the words contributing to the model's prediction. In 46 "situations" and "struggle" has the highest importance suggesting that if removed, the ADHD prediction would significantly change. The word "school" also plays a crucial role suggesting that discussions around struggling in school may be indicative of ADHD symptoms. Words like "staying," "social," and "focused" show a relationship with restlessness and hyper activity. They could also show persistent behavioral patterns. "Staying" may suggest that having trouble remaining in one place is a key ADHD trait. "Social" may indicate difficulties in social settings like school and "focused" could be having trouble staying attentive in class. "i" and "in" could be partial tokenization effects of the large language model. In conclusion, the SHAP analysis shows that words related to difficulty in social interactions and difficulty staying still are strongly related to ADHD prediction. Clinically, the words align with symptoms of hyperactivity and inattention.

X. MLOPS DEPLOYMENT

The model is deployed using Gradio, a user friendly interface that allows for interaction with machine learning models. It was used to build the web-based user interface where the users can interact with the model and receive their results in real time. The model was incorporated using a retrieval augmented generation pipeline with a cache store to store context, user responses, and questions asked by the chatbot. A gradio user interface was then built so as to allow easy user interaction. The application is then hosted onto a server to ensure accessibility removing the need to perform local installations of the necessary packages. In case of expansion, a docker can be used to contain dependencies and run the model in a standardised environment. Git hub is used for updating the versions. To maintain the reliability of the

model, model responses are monitored using gradio and issues such as incorrect predictions or contextually inaccurate responses are logged and stored for further analysis. Github manages different versions of the model while platforms like kaggle allow for the model to be reloaded and tweaks to be made using the freely available resources provided. Metrics scored such as F1 score, accuracy, and specificity, among others are periodically assessed to ensure continued reliability. System alerts could also be set up to accommodate for system failures. This would allow for them to be caught on to early and fixed before they cause strenuous problems. Challenges with MLOPS include scaling the model, handling multiple users efficiently requires a steady backend configuration meticulously designed with various rollbacks to prevent the system from crashing. Model performance may degrade as medical research is always being updated. What may not have been a symptom before could become one with time. The external corpus would require consistent updates to accommodate this. The model can also be retrained to incorporate newly founded data. Error handling so that the user interface gracefully handles incorrect inputs or unexpected behavior. Some best practices for the model would be auto-scaling, the model could be deployed with cloud services that support automative scaling to accommodate increase in usage. Retraining the model to as real world data on learning disabilities changes so that the model is able to keep up with the changing data and ask accurate, up-to-date questions. In addition, implementing authentication measures to the user interface so as to maintain user's privacy and protect their data. It is also important that there is compliance with data security and privacy laws to prevent encountering legalities that can lead to payments of heavy fines and sometimes even jail time. Some more good practices could be automating data-preprocessing, to save time. This would also mitigate the errors that may occur from human error. In addition, there can be the employment of real-time monitoring platforms like grafana to track the model's performance.

XI. LIMITATIONS

There were strict time constraints to get these models up and running. The limited amount of time available to train and test the models caused difficulties and made it impossible to build the best possible versions. Although the world is already aware of learning disabilities, there is more content available on some disabilities as compared to others. For example, while compiling data, it was realized that disorders like dyscalculia and auditory processing disorder were highly selected against. This made it difficult to come up with a diverse and unbiased dataset with little information available on these disorders.

Training the models is computationally expensive. The free graphical processing units made available by platforms like Kaggle and google collab can barely handle the weights of these models. They also employ time constraints on the time for which one has free access to resources. This introduces a speed factor at the rate at which someone has to work

thus making it difficult to maintain accuracy while building a model such as this, a medical chatbot that requires high levels of accuracy. Not only are the computational resources expensive, but acquiring the necessary access to the models takes a while. For example, access to pre-trained models like llama require authenticated access which can take days to acquire and still was not granted. This slowed down the rate of progress during construction of the model as necessary steps had to be taken to select a different model to fine-tune. Some models like GPT required hefty payments to train the model, fine-tune and implement RAG. Costs to perform all these tasks were charged separately making the model inaccessible to non-working students.

The lack of a fully compiled dataset to train conversational models on the detection of learning disabilities also proved to be a problem. This meant that data had to be collected manually, which was a difficult process as some sites had issues with being scraped for content. This caused frustration and also delayed the construction of the model. There was difficulty in constructing a large unbiased dataset with the resources available. Some clinical trials were closed-source, some texts in the field too. The gained access took a while to acquire, causing setbacks in the construction of the models. Different models have different prompt structures on which they can be trained and they understand. This meant constant restructuring of the dataset which was extremely time consuming. There is limited interest and support for applying machine learning techniques to medical diagnosis, resulting in outdated resources and limited guidance on structuring medical datasets and leveraging models effectively. This caused multiple errors during the set up and loading of the models and as a result caused frustration and still further delayed the construction of the models.

The inability to load the models directly also caused issues as quantized distilled models had to be used which although saved computational resources came with a compromise in accuracy and precision even after training. If the dataset was too complex, the model would struggle with providing proper responses that conformed to the defined prompt. With the necessary changes, the fine-tuned model tried to engage in a conversational dialogue with the user but ultimately crashed in-between tries. At first, while developing the RAG pipeline, the user symptom-based input was supposed to be used as input to query the pipeline and base the context retrieved on that. However, this proved to be a problem when the model asked questions that required simple answers like "yes or no". These could not be used to query the RAG pipeline. This meant that there had to be a cap on the number of times the user query could be used as input to index the vector database.

Balancing between all the performance metrics proved to be a problem as sometimes the model would not fit well to the data due to problems like class imbalance which made it difficult to maintain high performance metrics. Methods to test the models were limited as they were conversational and simulating human conversations of a child with learning

disabilities was quite difficult. This made it difficult to evaluate the performance metrics such as the accuracy, precision, F1 score, among others of the model. Handling disabilities; the chatbot correctly handled detecting all five disabilities. However, it failed at identifying cases where the disorders would overlap. This could raise some questions on its accuracy. If a child has more than one disability present, the chatbot will more likely determine one and fail to recognize the other. This can lead to cases of misdiagnosis. Although the models could predict the disabilities well, they failed at maintaining contextually relevant conversations with the users. This was especially true for the fine-tuned model. It struggled with making use of the context to ask relevant questions as sometimes it would ignore the context it was provided with completely. It also output some irrelevant information whenever prompted for a response from the user.

XII. CONCLUSIONS AND FUTURE WORK

The need for increased awareness of lesser-known learning disabilities is evident, as most research and resources focus primarily on widely recognized conditions like ADHD and dyslexia. In contrast, disabilities such as dyscalculia and APD lack sufficient research, making it difficult to construct comprehensive and unbiased datasets. The limited availability of information on these disorders necessitates collaboration with learning disability professionals to curate high-quality data. Additionally, in Uganda, few organizations prioritize learning disabilities, and those that do often focus only on the most common ones. Even globally recognized guidelines such as the DSM-5 lack detailed information on certain learning disabilities, further hindering data collection and diagnosis.

Model selection posed another challenge, as most state-of-the-art large language models require restricted access from their respective developers. This limits flexibility in experimentation and increases dependency on publicly available models. Different models are optimized for distinct tasks—some excel in mathematical computations, others in code generation, and some in text-based applications. Choosing the appropriate model requires a deep understanding of the intended application, dataset structure, and market availability. DeepSeek was selected due to its accessibility and compatibility with the chatbot's diagnostic requirements.

Evaluation metrics such as F1 score, precision, and accuracy are essential for assessing model performance, but they are insufficient on their own. Real-world testing remains the true measure of a model's effectiveness. Despite achieving high accuracy during testing, real-world deployment may reveal unforeseen limitations. The ultimate goal of this model was to generate diagnostic reports that psychologists could use as supplementary tools when diagnosing children with learning disabilities. The reliability of these reports, however, depended on the model's ability to accurately interpret user responses and detect disabilities without misclassification. The presence of overlapping symptoms created additional challenges, as the model sometimes failed to detect co-

occurring disabilities.

The dataset used for training was not large enough to fully fine-tune the model, highlighting the need for more diverse question-answer pairs. A more extensive dataset would allow the chatbot to generate more accurate responses by improving its understanding of varied symptom descriptions. Some models exhibited bias toward certain disabilities due to the unequal availability of data, further underscoring the necessity of RAG. The large external corpus used in the RAG pipeline helped compensate for dataset limitations by providing additional context. This significantly improved the chatbot's ability to ask more precise follow-up questions and engage in dynamic interactions with users. The incorporation of cache-augmented generation (CAG) further enhanced efficiency by reducing redundant context retrieval, making responses faster and more contextually relevant.

Moving forward, funding should be allocated to field research that collects real-world data, rather than relying solely on existing datasets. A progressive approach should be taken, where datasets for individual learning disabilities are expanded and tested separately before integrating them into a comprehensive model. This method would ensure balanced representation across all disabilities, including those that are currently underrepresented. Additionally, models should be fine-tuned in stages, gradually increasing dataset complexity to account for comorbidity—when multiple LDs coexist in a single child. While early diagnosis is critical, LDs are not always detected in childhood. Expanding datasets to include cases from adolescents and adults would improve inclusivity, allowing the chatbot to assist a broader population.

Beyond improving dataset diversity, incorporating multi-modal data sources would enhance diagnostic capabilities. For example, integrating handwriting analysis could improve the detection of dysgraphia and dyslexia, while speech recognition could be used to identify APD. Advanced image-processing techniques could enable the chatbot to analyze facial expressions and behavioral cues for autism diagnosis. These enhancements would extend the chatbot's functionality beyond text-based interactions, making it more robust in real-world diagnostic applications.

This model serves as a foundational step toward expanding machine learning applications in medical diagnostics, particularly in regions with limited access to trained psychologists. It offers a scalable alternative to traditional one-on-one evaluations by leveraging AI-driven conversations to identify potential learning disabilities. The integration of RAG and cache augmentation enables the chatbot to process complex symptoms efficiently and generate reports that bridge the gap between AI-generated insights and clinical application. By streamlining the diagnostic process, the system reduces the time and effort required for individualized assessments. Unlike many existing tools that focus on diagnosing a single disability, this model takes a more comprehensive approach, making it a valuable resource for early-stage diagnosis.

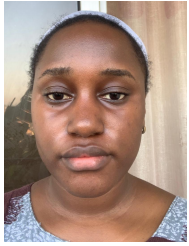
The development of this system also underscores the need for continued research into learning disabilities. Encourag-

ing data collection efforts and refining diagnostic models will contribute to the long-term goal of building a highly accurate and inclusive diagnostic tool. While the current model demonstrates promising results, future improvements should focus on refining conversational accuracy, enhancing dataset diversity, and ensuring that the chatbot effectively handles co-occurring disabilities. Expanding research efforts in this field will pave the way for more sophisticated AI-driven diagnostic systems capable of reaching underserved populations worldwide. In the future, the creation of larger and more structured datasets will further improve diagnostic accuracy. The inclusion of multimodal learning techniques such as integrating images and audio would allow for more comprehensive evaluations of LDs. For example, AI-powered handwriting analysis could enhance dysgraphia detection, and speech processing algorithms could improve the identification of APD. These advancements will allow the chatbot to function as a reliable and versatile diagnostic aid. Ultimately, the project lays the groundwork for groundbreaking research that could revolutionize the accessibility of learning disability diagnostics, ensuring that quick healthcare is available even in the most remote regions.

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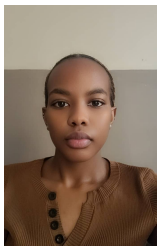
REFERENCES

- [1] R. Khan, H. Mittu, and D. Kumar, Dynamics of Learning Disabilities among School Students: A Review Analysis, pp. 20–38. 09 2024.
- [2] V. Galitskaya, A. Drigas, and -S. Antoniou, "Understanding the challenges of learning disabilities and psychosocial disorders," *Scientific Electronic Archives*, vol. 17, 10 2024.
- [3] M. Reuben, "Pretrained large language models and retrieval-augmented generation to address mental health challenges," 12 2024.
- [4] M. Laymouna, Y. Ma, D. Lessard, T. Schuster, K. Engler, and B. Lebouché, "Roles, users, benefits, and limitations of chatbots in health care: Rapid review," *J Med Internet Res*, vol. 26, p. e56930, Jul 2024.
- [5] Y. Ojo, O. Amuda, D. Emezirinwune, and V. Oluwabukunmi, "Integrating ai in mental health services: Challenges and innovations for information professionals," 12 2024.
- [6] K. P. Extension, "Leveraging technology for supporting students with learning disabilities," 06 2024.
- [7] S. Idrees, H. Thanganadar, W. Rehman, A. Mehmood, F. Azeez, and R. Mohammad, "Chatbots in healthcare – a study," *International Journal of Engineering Trends and Technology*, vol. 72, pp. 137–146, 07 2024.
- [8] S. Panjwani-Charania and X. Zhai, "Ai for students with learning disabilities: A systematic review," 09 2023.
- [9] M. K. Dobbala, "Conversational ai and chatbots: Enhancing user experience on websites," *American Journal of Computer Science and Technology*, vol. 7, 07 2024.
- [10] C. Jeong, "Domain-specialized llm: Financial fine-tuning and utilization method using mistral 7b," *Journal of Intelligence and Information Systems*, vol. 30, pp. 93–120, 03 2024.
- [11] S. Schickel, J. Welsch, M. Schweda, A. Hein, J. Rieger, and T. Kirste, "Ai-assisted ethics? considerations of ai simulation for the ethical assessment and design of assistive technologies," 04 2023.
- [12] C. Zakka, R. Shad, A. Chaurasia, A. R. Dalal, J. L. Kim, M. Moor, R. Fong, C. Phillips, K. Alexander, E. Ashley, et al., "Almanac—retrieval-augmented language models for clinical medicine," *NEJM AI*, vol. 1, no. 2, p. AIoa2300068, 2024.
- [13] S. Leon, F. Sunaira, S. Nehlin, s. A. M. Z. Abdal, and S. Momen, "A machine learning approach for early detection of learning disorders in pediatrics," pp. 1–6, 03 2024.
- [14] A. V. Ravindran, A. V. J., and M. P., "Prediction of learning disability using machine learning," August 2023.
- [15] V. Ramesh and R. Assaf, "Detecting autism spectrum disorders with machine learning models using speech transcripts," October 2021.
- [16] N. A. Chi, P. Washington, A. Kline, A. Husic, C. Hou, C. He, K. Dunlap, and D. Wall, "Classifying autism from crowdsourced semi-structured speech recordings: A machine learning approach," January 2022.
- [17] K. Lavanya, "Ai-driven adaptive learning: Enhancing education for students with disabilities," 2024.
- [18] S. Panjwani-Charania and X. Zhai, "Ai for students with learning disabilities: A systematic review," November 2023.
- [19] N. Ahire, S. Patkar, T. Prajapati, C. Patil, and V. Raj, "Identification of learning disability using ml," 2022.
- [20] H. Hollingsworth, "Ai is a game changer for students with disabilities. schools are still learning to harness it," December 2024.
- [21] G. Richard and M. Serrurier, "Dyslexia and dysgraphia prediction: A new machine learning approach," April 2020.
- [22] I. Dankwa-Mullan, "Health equity and ethical considerations in using artificial intelligence in public health and medicine," *Preventing chronic disease*, vol. 21, p. E64, 08 2024.
- [23] N. Amador, "Ethical implications of artificial intelligence in the healthcare sector," *Advances in Multidisciplinary and scientific Research Journal Publication*, vol. 36, pp. 1–12, 06 2023.
- [24] H. Lee, C. Wright, J. Ferranto, J. Buttner, C. Palmer, A. Welchman, K. Mazon, K. Fisher, D. Smelson, L. O'Connor, N. Fahey, and A. Soni, "Artificial intelligence conversational agents in mental health: Patients see potential, but prefer humans in the loop," *Frontiers in Psychiatry*, vol. 15, 01 2025.
- [25] P. K. Chenchala, A. Choppadandi, J. Kaur, S. Naguri, and R. Saoji, "Ethical considerations in the use of artificial intelligence and machine learning in health care: A comprehensive review," *Cureus*, vol. 16, 06 2024.
- [26] F. N. Lekota, "Governance considerations of adversarial attacks on ai systems," *International Conference on AI Research*, vol. 4, pp. 227–233, 12 2024.
- [27]
- [28] A. S. Miner, N. Shah, K. D. Bullock, B. A. Arnoff, J. Bailenson, and J. Hancock, "Key considerations for incorporating conversational ai in psychotherapy," *Frontiers in psychiatry*, vol. 10, p. 746, 2019.
- [29] T. Liu, S. Giorgi, A. Aich, A. Lahnama, B. Curtis, L. Ungar, and J. Sedoc, "The illusion of empathy: How ai chatbots shape conversation perception," 2024.
- [30] D. Manolescu, H. Alzu'bi, and E. Secco, "Interactive conversational ai with iot devices for enhanced human-robot interaction," *Journal of Intelligent Communication*, p. 2025, 01 2025.
- [31] H. Zhang, J. Song, J. Zhu, Y. Wu, T. Zhang, and C. Niu, "Rag-reward: Optimizing rag with reward modeling and rlhf," 01 2025.
- [32] P. Papalexandratou, A. Stathopoulou, and C. Skanavis, "The use of artificial intelligence in the education of students with learning disabilities -a systematic review," 01 2025.
- [33] G. Xiong, Q. Jin, Z. Lu, and A. Zhang, "Benchmarking retrieval-augmented generation for medicine," in *Findings of the Association for Computational Linguistics: ACL 2024* (L.-W. Ku, A. Martins, and V. Srikumar, eds.), (Bangkok, Thailand), pp. 6233–6251, Association for Computational Linguistics, Aug. 2024.
- [34] G. Xiong, Q. Jin, X. Wang, M. Zhang, Z. lu, and A. Zhang, "Improving retrieval-augmented generation in medicine with iterative follow-up questions," 08 2024.
- [35] A. A. Alanazi, "Understanding auditory processing disorder: A narrative review," *Saudi Journal of Medicine and Medical Sciences*, vol. 11, pp. 275–282, Oct-Dec 2023.
- [36] D. C. Lohani and B. Rana, "Adhd diagnosis using structural brain mri and personal characteristic data with machine learning framework," *Psychiatry Research: Neuroimaging*, vol. 334, p. 111689, 2023.



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